The Making of Dominant Vehicle Currencies: Evidence from DeFi

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Abstract

We use 472 million swaps to observe vehicle currencies in transaction-level routes on decentralised exchanges. Between matched January–June periods in 2024 and 2026, stablecoins' route-count share among native and stable vehicles rises from 16.9% to 42.3%. Substitutions offset within continuing endpoint pairs; reallocation toward stablecoin-intensive pairs and pair entry and exit generate the aggregate increase. Vehicle choice at pair entry predicts later choice. Following persistent bridge support, first-month use of the supported stablecoin is 42.6% below one-tenth of WETH depth and 84.1% at or above it. Trading relationships and capital scaling inside established pools jointly shape vehicle dominance.

1 Introduction

Every cross-border payment between two currencies needs an institution willing to receive one and deliver the other. When a payment provider neither participates in the destination payment system nor holds the destination currency, the BIS Project Nexus design assigns that task to a third-party foreign-exchange provider (Bank for International Settlements Innovation Hub, 2024). Project Rialto goes one step further: its experimental cross-border system uses a third currency when direct exchange is unavailable (Bank for International Settlements Innovation Hub, 2025). The practical question is which currency connects a thin trading relationship and what liquidity makes that connection viable.

The same problem underlies the dominance of the U.S. dollar. The dollar appears on more than 85% of foreign-exchange transactions, far more often than the United States' share of cross-border economic activity would suggest (Somogyi, 2026). Its use as a vehicle between other currencies concentrates trading in dollar markets; deeper markets lower trading costs and reinforce the dollar's intermediary role (Krugman, 1980; Kiyotaki and Wright, 1989; Flandreau and Jobst, 2009). This feedback explains persistence after a vehicle has become dominant. It says less about how a challenger gains share when new trading relationships and new market technologies appear.

Vehicle dominance depends on both the intermediary selected within a source–destination pair and the amount traded across different source–destination pairs. Aggregate dominance can therefore change even when the vehicle used inside every continuing pair remains unchanged. Measuring these two channels requires transactions that attach each intermediary to the exchange between the final source and destination.

Standard foreign-exchange records report turnover in each currency pair without linking the legs of the customer's exchange.[1] A yen–euro exchange routed through dollars appears as separate yen–dollar and dollar–euro trades. Somogyi (2026) infers vehicle activity from currency-pair volume responses around non-overlapping holidays. In decentralised exchange, by contrast, the pool calls inside one blockchain transaction preserve their execution order. We use these records to observe which asset connects the source to the destination.

We call the ordered source and destination the *endpoint pair*, and each pool trade a *leg*. A route may contain one leg, several sequential legs, or a split across pools. We reconstruct connected routes from 472 million swaps on eight Ethereum automated market maker (AMM) deployments between February 2020 and June 2026. The resulting token sequences identify the observed endpoint pair, the intermediary, and whether the route spans exchanges. They reveal vehicle use directly, avoiding inference from turnover in separate legs.

Stablecoins, including USDC, USDT, DAI, and other pegged tokens, become substantially more prominent vehicles during the sample. We compare the same January-through-June dates in 2024 and 2026 because the 2026 data end on June 30. Stablecoin dominance rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by dollar-weighted volume. Within endpoint pairs active in both periods, positive and negative stable-for-native substitutions offset in aggregate. The increase instead comes from trading shifting toward continuing pairs that already rely more on stablecoins and from pairs observed in only one comparison period.

The rotation is not a uniform move into every stablecoin. Native assets regain share in 2023 and 2024 before stablecoins recover the dollar-weighted lead in 2025. USDT's intermediary use rises especially sharply relative to its appearance at route endpoints. The stablecoin gain appears on routes confined to one exchange and on routes spanning several exchanges.

Entry does more than change the set of pairs. The vehicle used when an endpoint pair first appears strongly predicts the vehicle used over the next 120 days, including after removing pairs with WETH at an endpoint and controlling for entry size and route structure. Persistent bridge support also commonly precedes first use of the supported stablecoin. First-month adoption is 42.6% when stablecoin depth is below one-tenth of WETH depth and 84.1% once it reaches that threshold. The gap survives controls for prior WETH activity and remains among pairs with two non-stablecoin endpoints. Once both alternatives are used, relative deposited capital predicts which vehicle carries the trade. At first use, 92.5% of supported-stablecoin bottleneck capital lies in pools that were already active one week earlier, and capital in those continuing pools grows more than in the matched WETH bridge. These descriptive associations locate the vehicle role where pair formation and deposited capital meet; they do not identify an exogenous liquidity-supply effect.

Directly observed routes change the interpretation of vehicle-currency dominance. A large aggregate rotation can arise because the trading network expands around a challenger, even when

[1] For example, see the BIS 2025 Triennial Survey reporting guidelines and templates.

continuing pairs show little net substitution. This formation channel complements the holiday-based inference of Somogyi (2026) and the historical network evidence of Flandreau and Jobst (2009). The persistence and depth results connect that channel to classical theories in which vehicle use and liquidity reinforce one another, and to fragmented-market research in which liquidity supply determines the trading opportunities available to investors (Krugman, 1980; Chen and Duffie, 2021; Lehar and Parlour, 2025). A vehicle currency is made when it wins existing trades and when it becomes the liquid bridge inherited by new ones.

2 Setting and data

2.1 Routes and notation

Decentralised exchanges distribute liquidity across smart-contract pools that routers can combine atomically. A trader may use one pool, divide an order across several pools, or trade through an intermediary asset. Because the pool calls share a transaction, the intermediary sequence is observable.[2] A route through an intermediary is unavoidable when no direct pool connects the assets; when direct and indirect pools coexist, the router compares their fees, price impact, gas, and venue access. Lehar and Parlour (2025) describe the constant-product design and the economics of liquidity supply in these markets.

Blockchain logs record each pool call in execution order. We recover a route by following the directed token flows within a transaction. Calls with a common source and destination form a split order; a token received in one call and spent in a later call joins the calls into a sequential route. The same rule separates unrelated conversions bundled in one transaction. Throughout, for a reconstructed sequence $A \to B \to C$, we call (A, C) the ordered *endpoint pair* and each pool trade, $A \to B$ or $B \to C$, a *leg*. The order in (A, C) records the exchange direction, while the route records the intervening sequence $A \to B \to C$. Each leg trades in a liquidity pool; an endpoint pair need not have a direct pool. These endpoints belong to the connected pool component and need not coincide with the endpoints of a broader user instruction. We use *pair* as shorthand for endpoint pair only where no pool-level pair is under discussion.

A transaction on January 10, 2026 illustrates both the information in the logs and its limit. A public explorer describes a 460,000 PYUSD–USDe swap executed through 0x.[3] The user's instruction begins with PYUSD, but the connected pool sequence begins when 459,929.81 USDC enters Fluid. Fluid returns 460,331.05 USDT, which then enters Uniswap v4 and produces 460,104.28 USDe. The observed endpoint pair is therefore USDC–USDe, the two legs are USDC→USDT and USDT→USDe, and USDT is the vehicle. The earlier PYUSD–USDC conversion occurs outside the connected pool sequence, so the data do not show whether USDC was another vehicle or inventory supplied by the executor. We apply the connected-component rule to every transaction and do not infer an unobserved leg from nearby transfers. This choice may understate the intermediary role

[2] Ethereum records a transaction's ordered logs in its receipt; see the official Ethereum JSON-RPC documentation.

[3] Transaction `0xbda1d07f…9bf67ad9`.

of assets such as USDC when a broader instruction is funded outside the observed pool sequence.

The sequence directly identifies intermediary use. Conditional on reconstructed pool states immediately before execution, it also defines a same-size cost comparison between the observed path and available alternatives. We introduce notation for the route here and develop the cost comparison in Section 4.

Write $\mathcal{A}_t$ for the assets tradable on day t and $\mathcal{P}_t$ for the pools live on that day. For a route r, let $e(r)$ record its block, transaction position, and first pool-event position. The notation $e(r)^-$ refers to the state immediately before that transaction begins; a daily closing state generally occurs later. A route carries an ordered pool sequence $(p_1, \ldots, p_{m(r)})$ and token sequence $(k_0, \ldots, k_{m(r)})$, where $k_0 = i$ is the input and $k_{m(r)} = o$ is the output. Thus (i, o) is the endpoint pair and each (k_{h-1}, k_h) is one leg. The dollar input notional q corresponds to $\Delta_i(q, e) = q/P_{i,e}$ units of the input token. Let χ_{p,e^-} denote pool p's state immediately before execution and let $Q_p(i \to j, \Delta_i; \chi_{p,e^-})$ denote the units of token j returned for an exact input of Δ_i units of token i. Composing these pool quotes gives the route's token output and dollar value,

$$\begin{aligned} \Delta_{k_h} &= Q_{p_h}\left(k_{h-1} \to k_h, \Delta_{k_{h-1}}; \chi_{p_h, e(r)^-}\right), & h &= 1, \ldots, m(r), \\ Q_r(q, e(r)) &= \Delta_o, & O_r(q, e(r)) &= P_{o,e(r)} Q_r(q, e(r)). \end{aligned} \tag{1}$$

The venue-specific forms of Q_p appear in Appendix B. Pool invariants take token quantities as arguments, which is why the dollar notional q and the token input Δ_i remain distinct. Equation (1) describes an executed route and, once market state has been reconstructed, a hypothetical route.

Intermediary use is one of several currency roles. Invoicing concerns the currency named in trade contracts (Gopinath et al., 2020; Amiti et al., 2022; Mukhin, 2022); funding concerns the denomination of liabilities (Gopinath and Stein, 2021; Eren and Malamud, 2022); and dealer turnover records trading activity (Somogyi, 2026). These roles can reinforce one another. The path in Equation (1) identifies the asset used between the currency sold and the currency bought.

This route setting makes the classical volume–cost mechanism observable at transaction level. Greater use of a vehicle expands trading in the two markets joined through it; deeper markets can then lower the cost of using the same vehicle again (Krugman, 1980). In Kiyotaki and Wright (1989), traders accept an asset for indirect exchange when its marketability compensates for its storage cost. Dowd and Greenaway (1993) emphasise the fixed costs of learning, accounting, and quoting in a new currency. Liquidity placement, venue access, and the routes available to software provide corresponding coordination margins on decentralised exchanges.

2.2 Measuring vehicle dominance

An exact two-leg route consists of one pool trade on either side of a unique intermediary. This gives the primary measure a simple unit: each endpoint exchange contributes one intermediary observation. A supplementary multiple-intermediary measure includes routes of any length and counts each intermediary use separately. For route r, let $\mathcal{M}(r)$ denote the set of assets between its source and destination. We record the intermediary status of asset k as

$$Z_{r,k} \;=\; \mathbf{1}\{k \in \mathcal{M}(r)\}.$$

Let $a(k)$ assign each token to an economic asset type. The native group consolidates ETH and its one-for-one wrapped representation, WETH, after route reconstruction. The stable group contains fiat-reserve, on-chain-collateralised, synthetic, and non-dollar stable currencies. For $g \in \{\text{native}, \text{stable}\}$, define $Z_{r,g} = \sum_{k:a(k)=g} Z_{r,k}$; on an exact two-leg route, this sum is an indicator.

This grouping preserves a token-level estimand. WETH is the one-for-one token representation that allows ETH to interact with ERC-20 pool contracts; wrapping changes the technical form, not the underlying settlement exposure. We therefore reconstruct routes at the token level and consolidate ETH and WETH only when assigning the economic asset used as a vehicle. Dollar-pegged tokens remain distinct instruments because their issuers, redemption arrangements, risks, and leg-level liquidity differ. USDT in the USDC–USDT–USDe example is therefore an intermediary token even though all three instruments target the same unit of account. The aggregate stable group measures dominance by a family of stable tokens, while the token-level results retain issuer-specific competition.

We begin with the competition between native and stable intermediaries. Let $\mathcal{R}_t^{(2),NS}$ denote exact two-leg routes on day t whose intermediary belongs to either group, and let $\mathcal{R}_t^{(2),NS,V}$ retain routes for which the source, intermediary, and destination values agree within 20%. The equally weighted and dollar-weighted group shares are

$$S_{g,t}^{N} = \frac{\sum_{r\in\mathcal{R}_t^{(2),NS}} Z_{r,g}}{|\mathcal{R}_t^{(2),NS}|}, \qquad S_{g,t}^{V} = \frac{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r Z_{r,g}}{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r}, \tag{2}$$

where ν_r is the arithmetic mean of the route's total dollar value at its observed source and destination. The 20% rule also requires every intermediary's incoming and outgoing dollar values to have a ratio between 0.8 and 1.2. Thus $S_{\text{stable},t}^{N}$ is stablecoins' aggregate share among native and stable intermediaries, and $S_{\text{stable},t}^{V}$ is the corresponding dollar-weighted share. The two groups exhaust this denominator. Share levels are reported in percent; the unit for absolute share changes is one percentage point (pp).

The frequency with which an asset appears at the beginning or end of an observed pool route provides a benchmark for its intermediary use. This comparison uses a broader set of currencies $\mathcal{C}$: native, staked-native, stable, and imported currencies, with unclassified assets excluded. We retain every complete route that does not return to its starting currency. Direct routes contribute to the source and destination counts, and a longer route contributes one observation for each intermediary it uses. Let $n_{k,t}^{I}$ count intermediary uses and $n_{k,t}^{E}$ count source or destination appearances for $k \in \mathcal{C}$. The two shares are

$$I_{k,t}^{N} = \frac{n_{k,t}^{I}}{\sum_{j\in\mathcal{C}} n_{j,t}^{I}}, \qquad E_{k,t}^{N} = \frac{n_{k,t}^{E}}{\sum_{j\in\mathcal{C}} n_{j,t}^{E}},$$

and the corresponding share gap and excess-use ratio are

$$\Delta_{k,t}^{N} = I_{k,t}^{N} - E_{k,t}^{N}, \qquad X_{k,t}^{N} = \frac{I_{k,t}^{N}}{E_{k,t}^{N}} \quad \text{when } E_{k,t}^{N} > 0. \tag{3}$$

In Equation (3), a positive $\Delta_{k,t}^{N}$ means that the asset's share of intermediary use exceeds its share of source and destination appearances. We interpret $X_{k,t}^{N}$ as intermediation intensity relative to endpoint demand: zero means the asset is not used as an intermediary, one means its two shares are equal, and a value above one means it is overrepresented as a vehicle. The ratio is not a ranking of aggregate vehicle dominance. Dollar-weighted measures replace counts with route values while retaining the same currencies and the 20% agreement rule. Because this role-specialisation measure compares each currency's intermediary and endpoint uses, its denominator differs from the native-or-stable denominator in Equation (2). The latter remains the measure of aggregate rotation.

Source and destination assets must differ in the analysis sample because vehicle exchange connects two distinct endpoint assets. We classify routes that return to their starting asset as round trips and remove them. Such paths may contain cyclic arbitrage or other self-returning strategies: their economic object is a trading cycle, not an exchange from one endpoint asset into another, although trader intent remains unobserved. Daian et al. (2020) show that several arbitrage trades can be bundled atomically. On the median of 79 sampled days, round trips account for 12.7% of multi-leg routes by count and 21.7% by value. Appendix A reports their distribution across the sampled dates.

Within this route sample, we classify assets by their economic exposure: native, staked native, stable, imported, and other. The native category contains the platform's settlement asset. We treat ETH and its one-for-one wrapped representation, WETH, as one economic asset after reconstructing the route. This consolidation is an economic classification, not an address-level identity (Milionis et al., 2022). Liquid-staking tokens remain separate because they also convey staking returns. Stable assets target a fiat value, imported assets are wrapped representations of off-chain or cross-chain value, and the remaining assets enter the residual category; the asset's economic exposure determines its category. Stable assets themselves encompass economically different designs. Reserve-backed stablecoins rely on backing and redemption, making confidence in those arrangements relevant to acceptance and run risk (Gorton and Zhang, 2023); fiat-backed, crypto-collateralized, and algorithmic stablecoins also differ in their solvency and peg-restoration mechanisms (Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). Section 3.4 therefore examines the two largest reserve-backed stablecoins separately.

2.3 The route panel

The principal route panel covers Curve, Uniswap v2, v3 and v4, Balancer, SushiSwap v2 and v3, and Fluid on Ethereum. Uniswap v1 enters the separate protocol-architecture sample in Section 5.5. Its retained indexed records identify exchange contracts but omit their underlying token addresses, and the available files contain no complete exchange-to-token crosswalk. Shared transaction hashes and matching ETH legs still recover forced token-to-token routes, while their endpoint-pair identities remain unavailable for the principal panel. Table 1 summarises the principal panel's coverage. The sample spans 2,332 dates from February 11, 2020 through June 30, 2026. These deployments include several major exchanges but remain a subset of Ethereum trading. Protocols enter after their launch dates, so the exchanges and routes available to traders expand over the sample. We keep 55 early dates with no observed swaps as zero-activity dates for the covered deployments.

We begin with 472,254,909 pool-level swap records. Indexed blockchain data supply events for seven exchange series, Dune's `dex.trades` table supplies Fluid records, and direct Ethereum queries supply block headers, ordered logs and receipts, token precision, factory registries, and state checks.[4] After harmonising trade direction, event identity, and token units, the panel contains 471,616,269 legs. We order those legs within each transaction and apply the flow rule above to distinguish single-pool trades, split orders, sequential routes, and unrelated calls. Appendix A gives the reconstruction algorithm and reports the historical token units that we cannot verify.

Every reconstructed token sequence enters the count measure. Dollar-weighted results require source, intermediary, and destination values to agree within 20%. This is an eligibility test for a coherent dollar notional, not a control for route characteristics: covariates cannot repair a route whose token values do not reconcile. Keeping the count sample broader preserves observed intermediary choice when valuation is noisy; each value exhibit reports the stricter sample's coverage.

Transactions and contract-state changes are publicly observable (Schär, 2021). Reconstructing a route still requires matching pool events, venues, and token units. This procedure identifies the executed route and its calling contract, but an executor address does not establish which frontend or software formed the quote. Existing router labels likewise cover only part of decentralised-exchange activity (Xi and Moallemi, 2026). On a 2024 snapshot of 74,323 swaps, 241 distinct senders appear, and a registry of the ten largest accounts for 67.7% of those swaps. By a 2025 snapshot, 397 senders appear across 130,282 swaps and the same registry covers 11.8%. We therefore report route-level results without attributing them to a particular routing service.

Table 1: The route panel

	Route panel
Dates in the full route panel	2,332
Missing source-days	0
Raw swap rows	472,254,909
Usable directed route legs	471,616,269
Routes per day, median of 79 sampled days	161,152
Multi-leg routes per day, median of 79 sampled days	31,253
Multi-leg share of routes, median of 79 sampled days	19.7%
Round trips, share of multi-leg routes by count	12.7%
Round trips, share of multi-leg routes by value	21.7%

Note: The full eight-deployment panel contains UTC dates from 11 February 2020 through 30 June 2026. Raw swaps and usable legs use that full panel. The separate Uniswap v1 architecture sample lies outside these totals because its retained events do not recover endpoint-pair token identities uniformly. Daily route statistics and round-trip shares are validation quantities computed on 79 dates sampled across 1 June 2020 through 27 April 2026; this computational sample is not the support of the main dominance estimates. A multi-leg route traverses more than one pool; a round trip ends in the token with which it began. Round-trip shares are formed among multi-leg routes.

[4] See The Graph's indexing documentation, Dune's curated-data documentation, and the official Ethereum node-interface documentation.

Table 2: Stablecoin share among native and stable intermediaries

Stablecoin share among native and stable intermediaries	2024	2026	Change [pp] (s.e.)
Route count	16.9%	42.3%	+25.4 (1.05)
Dollar-weighted routes (20% agreement)	32.7%	76.5%	+43.9 (2.02)

Note: The denominator contains exact two-leg routes whose sole intermediary is native or stable. Daily shares receive equal weight over matched January–June dates in 2024 and 2026. Dollar-weighted estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth.

3 How stablecoins gain the vehicle role

3.1 The aggregate rotation

Stablecoins carry a much larger share of vehicle activity in 2026 than in 2024. The comparison uses the same January–June dates in both years because the 2026 panel ends on June 30. Stablecoins' share among native and stable intermediaries rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by routed value. Table 2 reports the paired-day estimates.

The annual series in Figure 1 shows that the change is not monotonic. Native assets regain share in 2023 and 2024; stablecoins recover the routed-value lead in 2025. The 20% value-agreement sample covers 71.6% of native intermediary value and 55.1% of stablecoin intermediary value in 2026. Fiat-reserve tokens account for +24.50 pp of the +25.42 pp count increase and +44.09 pp of the +43.87 pp value increase. The aggregate movement is therefore mainly a shift toward reserve-backed dollar tokens, although the token-level results below show that USDT and USDC do not move together.

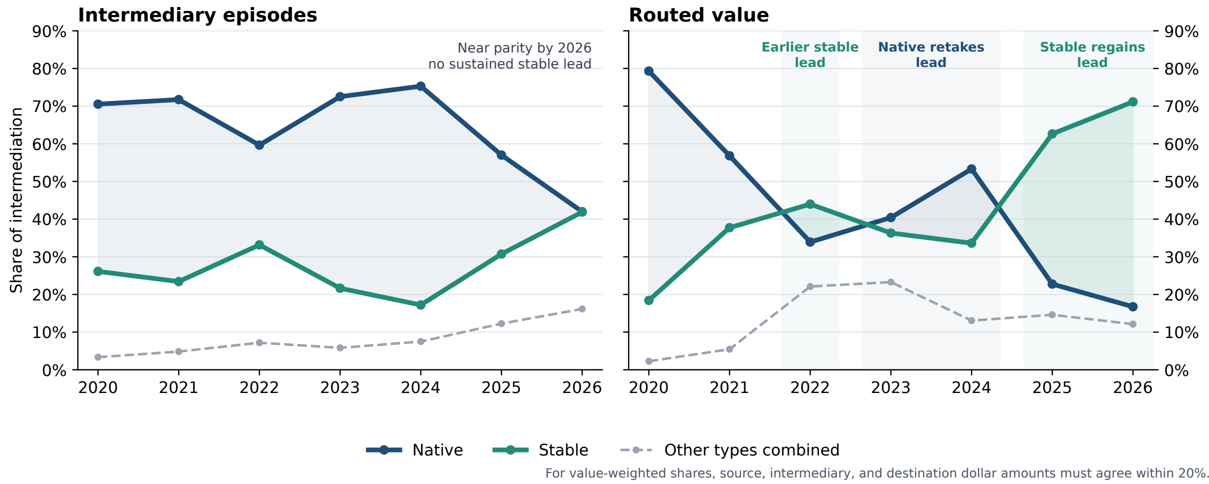
3.2 Endpoint-pair formation and persistence

Aggregate dominance can change inside continuing endpoint pairs or through the distribution of activity across pairs. Let p denote an ordered endpoint pair and $y \in \{2024, 2026\}$. A continuing pair carries native-or-stable routes in both matched periods; a year-specific pair carries them in only one. Let W_y be the share of native-plus-stable activity in continuing pairs, $\omega_{p,y}$ pair p's weight within that activity, and $s_{p,y}$ its stablecoin share. Continuing pairs have stablecoin share $S_{C,y} = \sum_p \omega_{p,y} s_{p,y}$, while year-specific pairs have share $S_{E,y}$. Writing $\Delta x = x_{2026} - x_{2024}$ and $\bar{x} = (x_{2024} + x_{2026})/2$ gives the exact midpoint decomposition

$$\Delta S = \underbrace{\bar{W} \sum_p \bar{\omega}_p \Delta s_p}_{\text{switching within continuing pairs}} + \underbrace{\bar{W} \sum_p \bar{s}_p \Delta \omega_p}_{\text{activity reallocation}} + \underbrace{(\bar{S}_C - \bar{S}_E)\Delta W}_{\text{continuing-pair weight}} + \underbrace{(1 - \bar{W})\Delta S_E}_{\text{year-specific pairs}} . \tag{4}$$

We also compare the same pair, date within the year, and realised single- or cross-exchange

Figure 1: Intermediary activity rotates between native assets and stablecoins



Note: The sample contains complete routes that do not return to their starting asset. Each route contributes once for every intermediary it uses, so routes with more than two legs can contribute several observations. Shares are formed across all intermediary types; Table 2 uses the narrower native-or-stable denominator. The 2026 observation covers January through June.

route class. For route-count or routed-value stablecoin share $S^{(m)}_{pds,y}$, the matched regression is

$$S^{(m)}_{pds,y} = \alpha^{(m)}_{pds} + \beta^{(m)}\mathbf{1}\{y = 2026\} + \varepsilon^{(m)}_{pds,y}, \qquad m \in \{N, V\}. \tag{5}$$

Here d is the month-day and s the realised route class. Observations are weighted by native-plus-stable route count or supported routed value; standard errors cluster by endpoint pair and date. The comparison is descriptive because route class and trading activity can respond to the same forces as vehicle choice.

Table 3 applies Equation (4) to route counts and supported value and reports Equation (5) on the matched pair-date-route-class cells.

The within-pair component is −0.1 pp by route count and 0.0 pp by routed value. These net estimates combine +1.3 pp from pairs moving toward stablecoins and −1.4 pp from pairs moving toward the native asset. Activity reallocation across continuing pairs contributes +8.6 pp by count and +26.2 pp by value, while year-specific pairs contribute +17.8 pp and +19.2 pp. Equation (5) reaches the same conclusion on the more restrictive matched sample: stablecoin share changes by +0.2 pp by count and −1.3 pp by value. A large aggregate rotation therefore coexists with little net switching inside established pair-date-route-class cells.

Endpoint demand sharpens the composition result. A pair with WETH at one endpoint cannot also use WETH as its intermediary, so the native-versus-stable choice is mechanically fixed. Such pairs supply +6.3 pp of the +8.6 pp count reallocation and +13.9 pp of the +21.0 pp entry component. An exact decomposition of the headline daily change by endpoint direction shows that pairs with one native and one stable endpoint contribute +9.2 pp of the +25.4 pp count increase

Table 3: Decomposition of the stablecoin rotation

Component or estimate	Estimate [pp]	Obs.
Panel A. Route-count share: market activity and vehicle incidence		
Endpoint pairs entering or leaving the sample	+9.8	
Endpoint pairs gaining or losing a vehicle route	−0.4	
Market activity shifting across continuing endpoint pairs	+7.9	
Change in how often continuing endpoint pairs use a vehicle	+7.0	
Stablecoin share within continuing vehicle-using endpoint pairs	+1.3	
Total route-count change	+25.7	
Panel B. Route-count share: continuing and year-specific endpoint pairs		
Net stablecoin-share change within continuing endpoint pairs	−0.1	
Pairs moving toward stablecoins (1,575)	+1.3	
Pairs moving toward native assets (1,489)	−1.4	
Vehicle activity shifting across continuing endpoint pairs	+8.6	
Weight of continuing versus year-specific endpoint pairs	−0.5	
Endpoint pairs traded in only one year	+17.8	
Total route-count change	+25.7	
Panel C. Dollar-weighted share: continuing and year-specific endpoint pairs		
Net stablecoin-share change within continuing endpoint pairs	0.0	
Pairs moving toward stablecoins (1,511)	+2.3	
Pairs moving toward native assets (1,445)	−2.4	
Vehicle activity shifting across continuing endpoint pairs	+26.2	
Weight of continuing versus year-specific endpoint pairs	−2.5	
Endpoint pairs traded in only one year	+19.2	
Total change in dollar-weighted share	+42.8	
Panel D. Matched ordered endpoint-pair estimates		
All two-leg routes, count share	+0.22 (0.76)	188,520
20% agreement sample, count share	+0.32 (0.75)	182,834
20% agreement sample, dollar-weighted share	−1.35 (2.19)	182,834

Note: Panels A through C compare pooled activity over matched January–June dates in 2024 and 2026. Panels B and C report Equation (4); their indented rows split the within-pair component into positive and negative contributions. Panel C weights routes by dollar value and applies the 20% agreement rule. Panel D estimates Equation (5); standard errors use two-way clustering by endpoint pair and date. Panels A and B factor the same route-count change in different ways, so their components should not be combined.

and +10.1 pp of the +43.9 pp value increase. Pairs with two stable endpoints contribute only +1.0 pp by count but +13.2 pp by value. Within that value channel, USDT alone contributes +13.7 pp; USDC, DAI, and the remaining stablecoin intermediaries together contribute −0.5 pp. The stable-to-stable result is present on all 181 common days, and trimming the largest decile of daily contributions leaves a +12.3 pp increase. The largest component still comes from all other endpoint pairs, at +15.2 pp by count and +20.6 pp by value. Stablecoin dominance therefore spreads beyond pairs whose endpoint identity limits the native vehicle, while routing between stable currencies concentrates on one intermediary issuer.

The first routes in a new pair predict its subsequent vehicle. For entrant pair p first observed on date t, let $S^{\text{entry}}_{p,t}$ be its initial stablecoin share and $S_{p,t+h}$ its stablecoin share through horizon h. We estimate

$$S_{p,t+h} = \alpha_h + \beta_h S^{\text{entry}}_{p,t} + \gamma_h \mathbf{1}\{S^{\text{entry}}_{p,t} > 1/2\} + X'_{p,t}\delta_h + \varepsilon_{p,t+h}, \tag{6}$$

where $X_{p,t}$ contains entry size, cohort year, endpoint class, direct-route share, and route-complexity share. The sample excludes WETH-endpoint pairs and requires complete follow-up support.

Table 4 shows that a 1 pp higher initial stablecoin share predicts +0.86 pp higher stablecoin share over 30 days and +0.74 pp over 120 days. The 120-day coefficient is +0.89 pp among pairs with no direct route at entry and +0.78 pp when routes are weighted by supported value. Persistence extends to issuer identity: a 1 pp higher entry share for a named stablecoin predicts +0.97 pp higher own-token share over 120 days. These estimates describe path dependence after pair formation; the initial vehicle is not randomly assigned.

3.3 Stablecoin arrival and network reach

Stablecoin arrival is concentrated among active native-only pairs. For an endpoint pair active on date t with no stablecoin vehicle, define $H_{p,t+30}$ as an indicator that a stablecoin route appears within 30 days. The rolling-hazard specification is

$$H_{p,t+30} = \alpha + \beta_N \log(1 + N_{p,t}) + \beta_A \log(1 + A_{p,t}) + \mu_d + \varepsilon_{p,t+30}, \tag{7}$$

where $N_{p,t}$ is origin-date route count, $A_{p,t}$ is pair age, and μ_d absorbs month-day effects. The mixed-vehicle analysis instead compares vehicles inside the same pair-date-route-class set and relates each vehicle's route share to its network reach outside that set.

Table 4: Endpoint-pair entry and later vehicle use

Model	Outcome	Regressor	Coefficient / (s.e.)	Obs. / clusters
Entry persistence	Stable follow-up share ($S_{p,t+h}$), 30 days	Entry stable share ($S_{p,t}^{\text{entry}}$)	$+0.86^{***}$ (0.06)	599,870 / 333
Entry persistence	Stable follow-up share ($S_{p,t+h}$), 120 days	Entry stable share ($S_{p,t}^{\text{entry}}$)	$+0.74^{***}$ (0.13)	512,201 / 243
Entry persistence	Stable-majority follow-up, 120 days	Stable-majority entry ($\mathbf{1}\{S_{p,t}^{\text{entry}} > 1/2\}$)	$+57.3^{***}$ (14.8)	512,201 / 243
No-direct-route split	Stable follow-up share ($S_{p,t+h}$), 120 days	Entry stable share ($S_{p,t}^{\text{entry}}$), no direct route	$+0.89^{***}$ (0.13)	512,020 / 243
Value-weighted entry	Stable value share, 120 days	Entry stable value share	$+0.78^{***}$ (0.14)	427,625 / 243
Named-stable identity	Own-stable follow-up share, 120 days	Entry named-stable share	$+0.97^{***}$ (0.02)	6,087 / 243
Secure-volume endpoint	Entry stable share ($S_{p,t}^{\text{entry}}$)	Stable endpoint × 2026	$+5.7^{***}$ (1.3)	615,333 / 363
Route architecture	Entry stable share ($S_{p,t}^{\text{entry}}$)	Direct-route share ($D_{p,t}$) × 2026	$+0.50^{***}$ (0.14)	615,333 / 363
Route architecture	Entry stable share ($S_{p,t}^{\text{entry}}$)	Complex-route share ($C_{p,t}$) × 2026	$+0.36^{***}$ (0.12)	615,333 / 363

Note: The entry-persistence rows estimate Equation (6). Coefficients appear above entry-date-clustered CR1 standard errors in parentheses. Share outcomes and share regressors are measured in percentage points (pp). The unit is a non-WETH entrant endpoint-pair-route-class observation except for the named-stable identity row, whose unit is a stablecoin entry observation with asset fixed effects. Rows are route-weighted except the stable-value row, which uses routed value satisfying the 20% agreement rule. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5 shows that one log point of origin-date route count predicts +3.5 pp higher probability of stablecoin arrival within 30 days, and pair age predicts another +1.6 pp. Once native and stable vehicles coexist in the same observed set, network reach predicts allocation: one log point of same-day reach is associated with +9.4 pp more route share, while prior-30-day reach predicts +7.8 pp. The formation evidence therefore links stablecoin entry to repeated pair demand and subsequent vehicle choice to the vehicle's wider trading network.

3.4 Issuer-level intermediation

USDC and USDT together account for 92.1% of the increase in stablecoin route-count share, but USDT supplies the clearer increase in role specialisation. Table 6 compares each token's share of intermediary use with its share of observed route endpoints. A ratio of zero means no intermediary use, one means proportional use in the two roles, and a value above one means the token is overrepresented as an intermediary.

On matched dates, USDT's intermediary share minus its endpoint share rises from −7.13 pp to +8.14 pp. Its intermediation-intensity ratio rises from 1.06 to 1.23 by count and from 0.59 to 1.42 by value. USDC changes little in the corresponding transition figure. The stablecoin rotation is therefore concentrated in one issuer's vehicle role, not a uniform consequence of stablecoin issuance.

Table 5: Stablecoin arrival and vehicle network reach

Model	Outcome	Regressor	Coefficient / (s.e.) [pp]	Obs. / clusters
Turn-on	Stable appears	Baseline route count	-2.7^{***} (0.4)	94,260 / 181
Turn-on	Stable appears	Direct-route share	$+27.1^{***}$ (6.4)	94,260 / 181
Turn-on	Stable appears	Complex-route share	$+20.3^{***}$ (5.1)	94,260 / 181
Thin/direct	Stable appears	Direct share × thinness	$+3.1^{***}$ (0.9)	94,260 / 181
Leader switch	Stable becomes leader	Baseline route count	-1.1^{***} (0.2)	94,260 / 181
Rolling hazard	Stable appears within 30d	Origin route count	$+3.5^{***}$ (0.7)	1,690,478 / 303
Rolling hazard	Stable appears within 30d	Endpoint-pair age	$+1.6^{***}$ (0.2)	1,690,478 / 303
Mixed risk set	Vehicle route share	Same-day reach	$+9.4^{***}$ (2.9)	27,852 / 363
Mixed risk set	Vehicle route share	Prior-30d reach	$+7.8^{***}$ (2.5)	27,852 / 363
Issuer split	Vehicle route share	USDC × 2026	$+53.0^{**}$ (22.5)	27,852 / 363
Issuer split	Vehicle route share	USDT × 2026	$+88.2^{***}$ (24.2)	27,852 / 363

Note: Rolling-hazard rows estimate Equation (7) on active native-only endpoint-pair dates with complete 30-day support. Turn-on and leader-switch rows compare matched January–June pair-date-route-class observations in 2024 and 2026. Mixed-set rows compare native and stable vehicles within the same observed pair-date-route-class set. Coefficients appear above clustered standard errors in parentheses and are reported in pp. Reach, route count, and pair age enter as $\log(1+x)$. The designs use month-day or risk-set fixed effects as appropriate and cluster by endpoint pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates are predictive associations.

Table 6: USDT intermediation relative to endpoint demand

	2024	2026
Count excess-use ratio (2024 full year; 2026 January–June)	1.06	1.23
Value-weighted excess-use ratio (2024 full year; 2026 January–June)	0.59	1.42
Paired January–June intermediary minus route-endpoint share [pp]	−7.13	+8.14

Note: Intermediary and endpoint shares are formed across native, staked-native, stable, and imported currencies on complete routes that do not return to their starting asset. Direct routes enter the endpoint denominator. Intermediation intensity is intermediary share divided by endpoint share; the gap is their difference. Ratios use full-year 2024 and January–June 2026, while the share gap uses matched January–June dates. Dollar-weighted measures apply the 20% value-agreement rule.

3.5 Endpoint demand within a day

The issuer comparison can be made within each date. Let $I_{a,t}$ be currency a's share of the day's intermediary episodes and $D_{a,t}$ its share of route-endpoint appearances, both in pp. We estimate

$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}_a' \theta + \mu_t + \eta_a + \varepsilon_{a,t}, \tag{8}$$

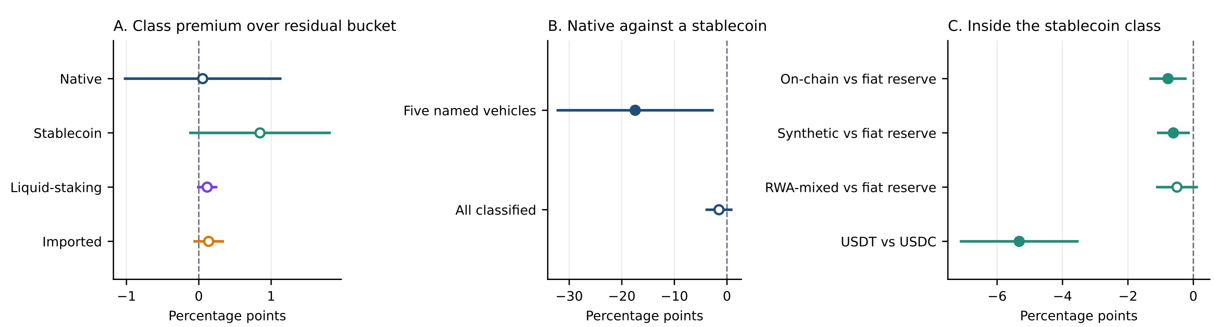
where $\mathbf{A}_a$ contains asset-class indicators, μ_t date effects, and η_a currency effects.

Table 7 reports a demand coefficient of 1.587 across currencies within a date and 0.985 within a currency after absorbing date and currency effects. The native-class coefficient falls from +34.55 pp to +0.05 pp when endpoint demand enters; the stablecoin coefficient remains +0.85 pp. Intermediary use therefore moves almost one for one with a currency's own endpoint demand, while stablecoins retain a small additional association. The leave-own-out denominator, alternative activity floors, and early-versus-late sample splits produce similar demand slopes; Figure 2 reports the class and issuer contrasts.

3.6 Exchange span

Figure 3 shows that cross-exchange routes grow from 2.0% of route count in 2020 to 57.2% in 2026 and account for 79.1% of routed value in 2026. Intermediation becomes less common over the same period, yet stablecoin share rises among the indirect routes that remain. Within a date, cross-exchange routes carry +14.18 pp more stablecoin share by routed value than single-exchange routes; the count difference is +3.73 pp. Exchange span is selected by the router, so the comparison describes where stablecoin-intermediated value travels and does not identify the effect of widening venue access.

Figure 2: Within-day intermediary role, conditional on endpoint demand



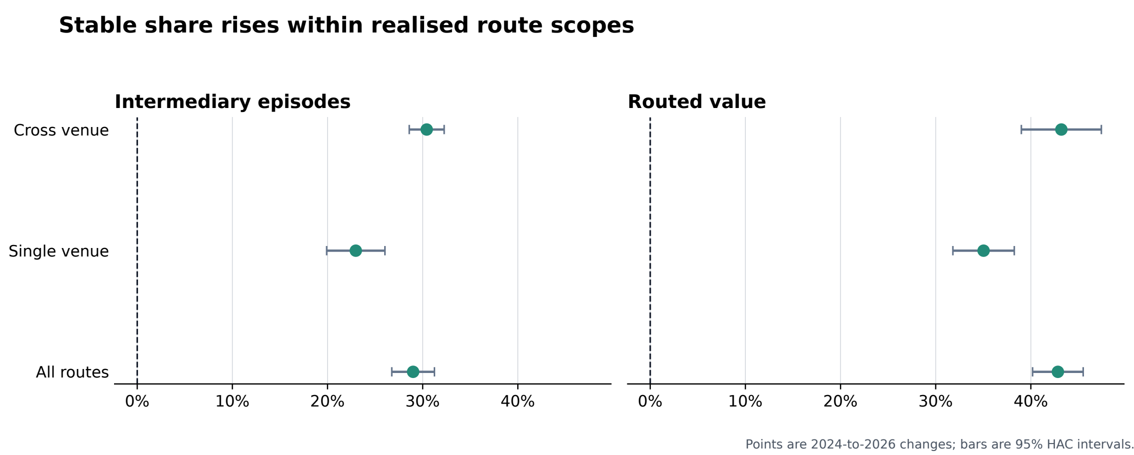
Note: Each row is one coefficient from Equation (8) with date fixed effects. Bars are 95% confidence intervals. Panel A compares asset classes with the residual unclassified group, Panel B compares the native asset with stablecoins, and Panel C compares stablecoin backing regimes and USDT with USDC. The panels use separate horizontal scales.

Table 7: Intermediary role within a day

	Intermediary episode share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	+34.56	+34.55	+0.05	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	+2.42*	+2.42*	+0.85*	—
	(1.32)	(1.32)	(0.50)	—
Own endpoint-demand share ($D_{a,t}$)	—	—	+1.59***	+0.98***
	—	—	(0.03)	(0.30)
R^2	0.485	0.485	0.953	0.208
Observations	1,828,862	1,828,862	1,828,862	1,828,862
Currencies	304,572	304,572	304,572	304,572
Dates	2,259	2,259	2,259	2,259
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

Note: The table estimates Equation (8) on 1,828,862 currency-days over 304,572 currencies and 2,259 dates. The outcome and demand regressor are measured in pp. Coefficients appear above standard errors in parentheses clustered by date and currency. The final column absorbs currency and date effects jointly, leaving class indicators unidentified. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure 3: Stablecoin intermediary share rises within single- and cross-exchange routes



Note: The sample contains complete routes that do not return to their starting asset, including routes with more than two legs. Each route contributes once for every intermediary it uses. Stablecoin shares use all intermediary types as the denominator. Count and dollar-weighted estimates compare matched January–June dates in 2024 and 2026. Dollar-weighted measures apply the 20% value-agreement rule.

4 Comparing direct and intermediated execution

Realised routes identify the assets that carried each trade. Execution cost is a separate object: it compares those routes with alternatives available immediately beforehand, quoted at the same trade size and over a common set of exchanges. We hold the source i, destination o, dollar input q, and pre-transaction pool state e^- fixed. The best direct and two-leg outputs through intermediary k are

$$O^D_{i,o,q,e} = P_{o,e} \max_{p \in \mathcal{P}_{i,o,e^-}} Q_p\big(i \to o, \Delta_i(q,e); \chi_{p,e^-}\big)\,,$$

$$O^I_{i,o,k,q,e} = P_{o,e} \max_{p' \in \mathcal{P}_{k,o,e^-}} Q_{p'}\left(k \to o, \max_{p \in \mathcal{P}_{i,k,e^-}} Q_p\big(i \to k, \Delta_i(q,e); \chi_{p,e^-}\big)\,; \chi_{p',e^-}\right)$$

This comparison extends the realised-versus-optimised allocation benchmark of Xi and Moallemi (2026) from parallel pools for one atomic asset pair to sequential routes through an intermediary.

The vehicle-route shortfall relative to direct execution is

$$\Delta C^D_{i,o,k,q,e} = \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}},$$

reported in basis points as $10^4 \Delta C^D_{i,o,k,q,e}$. Positive values mean that the direct route returns more output before gas. Swaps and liquidity changes are replayed in on-chain order because a daily or hourly closing state incorporates events that occurred after the trade. The observed event marks settlement; a quote formed earlier can reflect a different opportunity set.

We compare hypothetical legs only within a common executable range. Let $\widetilde{Q}_\ell$ denote leg ℓ's output at the pool's marginal rate. Each leg satisfies

$$\pi_\ell(\Delta_i, e^-) = \frac{\widetilde{Q}_\ell(\Delta_i, e^-) - Q_\ell(\Delta_i, e^-)}{\widetilde{Q}_\ell(\Delta_i, e^-)} \le \bar{\pi}, \qquad \bar{\pi} = 0.05. \tag{9}$$

The 5% own-price-impact ceiling applies across pool families; contract-specific limits add further restrictions. Appendix B gives the pricing equations and Appendix D calibrates the support rule.

Receipt gas measures the fixed execution toll attached to adding a vehicle leg. That toll matters because gas fees can change both trading costs and liquidity concentration in decentralised exchange (Caparros et al., 2024). In the executed-route gas sample, a stable-vehicle route in 2024 uses 94,290 more gas than the median direct route, 50.2% of the direct-route median. Evaluated at that year's median gas price and WETH price, the extra toll is \$3.24, so a trade needs about \$32.4k of notional before the extra gas is below one basis point. In 2026 the median stable-vehicle route uses 233,640 more gas than the direct-route median, but the dollar toll falls to \$0.08 and the one-basis-point notional falls to \$762. Pricing the same fixed toll at each stable-vehicle route's own gas price sharpens the interpretation: median stable-vehicle notional falls from \$832 to \$82, while the median fixed toll falls from 44.3 bp to 16.2 bp and the share of stable-vehicle routes with a fixed toll below 25 basis points rises from 39.5% to 55.7%, the trade-size version of the fixed-cost channel in Caparros et al. (2024). Thus the extra-hop overhead remains visible in gas units while becoming economically small for ordinary routed notionals. This evidence supports the interpretation that vehicle formation spread to routes for which fixed execution overhead had ceased to be a large barrier. The estimand is the executed route's fixed gas toll; a stable-versus-WETH all-in price ranking remains outside this check.

The benchmark represents public on-chain liquidity in the declared exchange set. Private order flow, financing costs, losses from reverted transactions, and pools outside that set can alter a router's choice. Pool-family coverage is economically material. In the sampled Curve exercise, pools outside the StableSwap model carry 34.2% of assessed volume, including 65.2% of native-leg volume and 21.1% of stablecoin-leg volume. Hook-bearing or dynamic-fee Uniswap v4 pools, SushiSwap v3, and several Balancer families also use pricing states outside the benchmark. Appendix E quantifies these boundaries.

5 What can account for the rotation?

The allocation results sharpen the mechanism question. Stablecoin dominance rose chiefly as routed activity moved across source–destination endpoint pairs and the incidence of intermediation changed, while net replacement of a native intermediary within continuing endpoint pairs contributed little. Any explanation for the rotation must therefore account for the assets being exchanged as well as the intermediary carrying the route.

5.1 Path complexity and exchange span

A competing account assigns the rotation to how routes are assembled. Paths grew longer and reached across more exchanges over the sample, and long paths already favoured stablecoins: in 2024 stablecoins carried 44.5% of native-plus-stable intermediary episodes on routes of more than two legs, against 16.9% on two-leg routes. That account predicts a roughly flat stablecoin share inside each complexity class and an aggregate increase produced by activity moving toward longer routes that were already stablecoin-intensive.

Stablecoin share rises within every complexity class. Among intermediary episodes on two-leg routes, the stablecoin share rises +25.4 pp by count, with a standard error of 1.05 pp, and +43.9 pp by value (2.02 pp). Among episodes on routes of more than two legs it rises +16.3 pp (1.86 pp) and +35.9 pp (1.68 pp), from a base of 44.5% to 60.7% by count. Both classes move, and the class with the shortest paths moves further, which leaves reweighting toward complex routes too small to supply the aggregate change. Dividing each complexity class again by whether the route touched one exchange or several leaves four groups, and every one of them rises; the weakest is +17.0 pp by count (1.36 pp) and +25.6 pp by value (1.52 pp). The stablecoin gain is thus common to short and long paths and to routes that stay on one exchange and cross-exchange routes.

Two boundaries limit what this comparison licenses. The number of legs records the path a trade actually took, so it measures construction; extra legs have no ordinal interpretation for execution quality. Complexity and exchange span are also jointly selected outcomes, since the same conditions that make a longer or wider path attractive can bear on the choice of intermediary. The comparison locates the stablecoin increase within each route-complexity and exchange-span class. The effects of routing software and exchange span remain unidentified.

5.2 The software that assembles the path

Traders on these exchanges seldom assemble a path by hand. A router contract holds the pool set, searches it, and returns a sequence of legs, and the routers in widest use over this sample were rewritten three times: the Auto Router in September 2021, its cross-version successor that December, and the Universal Router in November 2022.[5] Each release widened the set of paths an ordinary trader could reach without extra effort. The alternative this raises is that the intermediary is a residue of the search: the software found a path through a stablecoin because such paths existed and had become cheap to look for, so the rotation would record an improvement in search and not a change in what traders were willing to hold between legs. Search does leave room to move. On parallel pools for a single leg, realised routes fall measurably short of the optimum available in the same pool state (Xi and Moallemi, 2026), so a router that closes part of that gap changes which paths trades actually take.

The market-microstructure design used for staggered trading technologies is unavailable here. That literature identifies the effect of an exchange technology when the exchange grants it to some securities before others, because the securities still waiting absorb whatever else is moving in the market at the same time. A router release admits no such comparison. The contract is deployed once and is available immediately for every token pair, at every exchange the router indexes, to every trader, so there is no untreated group and no difference to take. What a symmetric window around each release can measure is whether the structure of routes moved at all when the search over them was rewritten, and that weaker object is the one the alternative needs. If automated search produced the intermediary, the incidence of intermediation should step up when the search was automated.

Route composition remains stable around the releases. Table 8 reports route composition over the 60 days on either side of each release, excluding the release date itself, on the daily calendar used throughout the paper. The share of economic routes carrying a third asset between the endpoints falls by -5.7 pp at the first release, rises by $+0.8$ pp at the second, and falls by -0.7 pp at the third, moving from 20.8% before the first release to 12.5% after the last. The largest increase at any of the three is $+0.8$ pp. Path length is as still: the mean number of legs on an indirect route moves by at most 0.049 at any release, and the share of indirect routes running past two legs changes by a few pp in either direction. Indirect routing as a whole behaves the same way, falling sharply at the first release and moving by less than half a pp at the other two.

[5] Uniswap Labs, *Introducing Auto Router*, September 16, 2021, and *Introducing Permit2 and Universal Router*, November 17, 2022. The cross-version release is dated from Uniswap Labs' v2-versus-v3 liquidity-provider study of December 16, 2021.

Table 8: Route structure around public router releases

	Indirect	Intermediated	Cross-exchange	Legs	Exchanges	Over two legs
Panel A: Auto Router, September 16, 2021 (60 days each side)						
Before	22.7	20.8	8.5	2.08	1.10	6.6
After	18.9	15.1	11.0	2.10	1.11	8.1
Change	−3.7	−5.7	+2.5	+0.01	+0.01	+1.5
Panel B: Cross-version Auto Router, December 16, 2021 (60 days each side)						
Before	18.5	13.9	11.3	2.11	1.11	8.4
After	18.6	14.8	14.4	2.16	1.16	11.3
Change	+0.2	+0.8	+3.1	+0.05	+0.05	+2.9
Panel C: Universal Router, November 17, 2022 (60 days each side)						
Before	19.8	13.2	26.4	2.19	1.22	13.1
After	19.5	12.5	24.4	2.20	1.20	12.0
Change	−0.3	−0.7	−2.0	+0.01	−0.03	−1.1

Note: Symmetric windows of 60 calendar days either side of each release, excluding the release date. Indirect is the share of economic routes with more than one leg; Intermediated is the share carrying a third asset between the endpoints; Cross-exchange is the share of intermediated routes touching more than one exchange; Over two legs is the share of indirect routes with more than two legs; Legs and Exchanges are means per indirect route. Shares and their changes are in pp. Economic routes exclude canonical endpoint round trips. The windows are market-wide and carry no split by intermediary currency. A release is available to every trader at once, so these are descriptive route compositions.

One margin does move, and it is the margin a router should move. Among intermediated routes, the share touching more than one exchange rises by +2.5 pp at the first release from a base of 8.5% and by +3.1 pp at the second; at the third it changes by −2.0 pp, from a level near a quarter of intermediated routes to 24.4%. Automated search integrated the exchange set without sending trades through more assets. The distinction is the one that governs this paper's measure, which counts the asset standing between the endpoints and not the number of venues its legs visit, and it is the same separation the exchange-span stratification of Section 5.1 makes within a single date.

Router releases expand exchange span while intermediary incidence remains stable. Three features of the windows bound that comparison. The first two windows overlap: the later window of the first release and the earlier window of the second share thirty days, so the first two comparisons are drawn partly from the same trading. The third window also contains a major market event. The earlier window of the third release contains the November 2022 failure of a large centralised exchange, and the composition change recorded there belongs at least as much to that event as to the router. A release date, finally, marks when a contract became callable while trader adoption can begin later, so a window straddling gradual adoption understates a movement that arrived slowly.

5.3 The pricing rule at each leg

The third alternative centers on the exchanges that price each leg, because AMMs apply different pricing rules across pools. The constant-product rule sets a price from pool reserves alone and moves it with trade size relative to depth, so it charges a trade for the price impact it causes regardless of how closely the two assets substitute. Curve's invariant instead interpolates toward a flat curve in the neighbourhood of parity, which is exactly where two dollar-pegged assets trade, and Appendix B.3 states the interpolation the venue applies. On this account the rotation is an adoption of technology: a leg through a stablecoin became cheap once venues priced pegged assets on a curve built for them, and the vehicle role passed to stablecoins because of where their legs could be priced, so no monetary property of the asset need enter the account at all. The timing is favourable to it, since stable-specialised pools grew over the same years as the rotation.

The account can be tested by restricting attention to routes on which the pricing rule never changed. A route component enters the principal panel's constant-product scope only when every one of its legs executes at Uniswap v2 through v4 or at SushiSwap v2 or v3, each of which prices locally on the product of reserves, concentrated liquidity inside an active tick included. No leg inside that scope can have been priced by a curve specialised for pegged assets, so a technology account predicts that stablecoins gain nothing there. The quantity compared is the excess-use ratio of Section 3.4, an asset's share of intermediary activity divided by its share of route-endpoint demand, formed separately inside each scope, so that the denominator for a venue family is the endpoint demand observed in that same family.

The constant-product restriction produces an even larger rotation. Where every leg is priced by the reserve product, stablecoin excess use rises from 0.78 in 2024 to 1.33 in 2026 by value, a movement of +0.55 against +0.40 over all venues, and the ratio crosses one inside the restricted scope in the same comparison. Native assets fall to 0.62 there over the same years. By count, where stablecoins were already over-represented as intermediaries at the start of the sample, the restricted and unrestricted paths are almost the same: 1.28 to 1.42 inside the scope against 1.27 to 1.41 over all venues. The scope also carries substantial activity, since routes priced end to end by the constant-product rule carry 84.8% of vehicle-currency intermediary episodes in 2026. Table 9 reports every year and every venue family.

The venue that embodies the alternative supplies no intermediation at all. In each year of the sample, route components all of whose legs execute on Curve contain no intermediary episode whatever. The stable-specialised curve is used to exchange two pegged assets with one another, not as a place through which a route passes on its way somewhere else, so the pricing rule that makes stablecoin legs cheap operates on the direct trades that the vehicle measure excludes by construction. Curve legs do appear inside routes that also touch another exchange, and those routes are counted in the all-venue scope; what the sample contains no instance of is a route intermediated entirely within the specialised invariant. Balancer, whose pools mix weighted and stable families and which the panel therefore leaves technologically unlabelled, carries 39,536 intermediary episodes in its busiest year against 8,767,213 in the busiest year of the constant-product scope, and its ratios swing accordingly; it is reported solely to delimit the sample's composition.

What the comparison licenses is again bounded. Holding the pricing rule fixed supplies only a descriptive restriction: venues are chosen alongside routes, and a trader who would have used

Table 9: Vehicle excess use by venue pricing family

	All venues		Constant product		Balancer		Curve	
Year	Count	Value	Count	Value	Count	Value	Count	Value
Panel A: Stablecoins								
2020	1.45	0.42	1.50	0.42	no routes		no intermediation	
2021	1.18	0.89	1.20	0.92	2.09	1.43	no intermediation	
2022	1.46	1.08	1.50	1.12	1.91	1.00	no intermediation	
2023	1.89	0.94	1.95	1.00	1.46	0.37	no intermediation	
2024	1.27	0.84	1.28	0.78	1.01	0.27	no intermediation	
2025	1.23	1.21	1.23	1.20	0.70	0.09	no intermediation	
2026	1.41	1.24	1.42	1.33	1.75	1.68	no intermediation	
Panel B: Native asset								
2020	0.91	1.56	0.90	1.37	no routes		no intermediation	
2021	0.94	1.13	0.94	1.09	0.24	0.66	no intermediation	
2022	0.85	0.96	0.85	0.93	0.35	0.90	no intermediation	
2023	0.87	1.16	0.88	1.08	0.50	0.53	no intermediation	
2024	0.95	1.20	0.95	1.29	0.22	0.20	no intermediation	
2025	0.89	0.67	0.92	0.85	0.19	0.19	no intermediation	
2026	0.77	0.58	0.80	0.62	0.51	1.30	no intermediation	

Note: Excess use is an asset type's share of intermediary activity divided by its share of route-endpoint demand, formed over the vehicle currencies within each scope. A route component enters a scope only when every one of its legs executes at a venue in that family. The principal panel's constant-product family is Uniswap v2 through v4 and SushiSwap v2 and v3. Uniswap v1 enters the separate forced-routing architecture sample. Balancer is listed separately and left technologically unlabelled because its pools mix weighted and stable families. Value ratios use all routes; the 20% value-agreement support of Table 2 is narrower. Balancer has no route components before 2021, and 2026 covers January through June.

a stable-specialised pool may appear in the constant-product scope only because that pool was unavailable for the leg at hand. The ratio's denominator removes the most obvious composition problem, since each scope is judged against the endpoint demand observed inside it, but selection into scopes remains. The rotation therefore persists beyond stable-specialised pricing: it is present, large, and in the same direction where that pricing was never applied.

5.4 Liquidity in the vehicle's own pools

The classical vehicle-currency mechanism links trading volume to lower transaction costs (Krugman, 1980). On a decentralised exchange, the relevant markets are the pools connecting an intermediary to other assets. Trading can attract provider capital, deeper pools can lower price impact, and lower price impact can draw further trading. Pool size remains an equilibrium outcome because providers balance fee income against adverse-selection risk and the cost of adjusting their positions (Lehar and Parlour, 2025). Deposited capital measures funds committed to a pool; a size-specific quote measures how much liquidity is executable within a stated price impact. The distinction matters. Concentrated-liquidity providers can alter executable depth by changing where capital is placed along the price curve, so deposited capital and executable liquidity coincide only under particular pool designs (Adams et al., 2021).

That mechanism operates through the two pool venues joining an intermediary to the endpoints. Deeper stablecoin pools can lower the price impact of carrying an endpoint exchange through a stablecoin, so the classical account asks whether flow moves away from the incumbent intermediary after a stable bridge becomes feasible. Section 3.2 conditions on realised routes in both endpoint years: within those matched endpoint pairs, stablecoin share changes by +0.2 pp by count, with an interval that excludes increases beyond +1.7 pp. Bridge feasibility and adoption remain pooled in that estimate. A separate event design therefore dates the feasible-set expansion before asking how the realised route changes.

Let event e identify an ordered endpoint pair (i, j), route scope, and the first day T_e on which DAI, USDC, or USDT has positive prior-calendar deposited capital on both route legs, with two-leg support present on at least 24 of the next 30 calendar days. The sample contains endpoint pairs that traded through WETH before T_e, have no earlier observed stablecoin route, and trade on at least three days in the prior month and each follow-up window. For route counts and supported value $q \in \{R, V\}$, the paired change is

$$\Delta S^q_{e,h} = S^q_{e,h} - S^q_{e,[-30,-1]},$$

where h is either the first 30 days or days 30 through 119 after bridge establishment. Share changes use the harmonic mean of pre- and post-event mass; route-intensity changes weight each event equally. Standard errors cluster by ordered endpoint pair and event date.

Panel A of Table 10 records the change around first persistent support. Across 865 events, stable route share rises by +7.93 pp in the first month and supported-value share by +3.11 pp. Native-route intensity changes by −0.20 log points, while total route intensity changes by −0.10 log points. The count-share increase over days 30 through 119 is +8.68 pp; the supported-value estimate is +6.94 pp. These event-window estimates date stable support and the associated route allocation. The observed support event is an equilibrium outcome.

The continuous-depth design measures the capital available on both routes after that support appears. Let $K_{av,t-1}$ be exact prior-calendar deposited capital supporting the leg from endpoint a to vehicle v, and let $\mathcal{S} = \{\mathrm{DAI}, \mathrm{USDC}, \mathrm{USDT}\}$. Define the best stable weak-leg depth D^S_{et}, WETH weak-leg depth D^W_{et}, their stable relative-depth share Q_{et}, and stable route-count share S^R_{et} by

$$\begin{aligned} D^S_{et} &= \max_{v \in \mathcal{S}} \min\{K_{iv,t-1}, K_{vj,t-1}\}, \qquad D^W_{et} = \min\{K_{iW,t-1}, K_{Wj,t-1}\}, \\ Q_{et} &= \frac{D^S_{et}}{D^S_{et} + D^W_{et}}, \qquad S^R_{et} = \beta Q_{et} + \alpha_e + \lambda_{m(t)} + \eta_{a(t)} + \varepsilon_{et}. \end{aligned} \tag{10}$$

Here W denotes WETH, $m(t)$ is calendar month, and $a(t)$ groups event age into seven-day bins. Equation (10) absorbs bridge-event fixed effects, weights by the event day's WETH-plus-stable route count, and clusters standard errors by ordered endpoint pair and date.

Panel B of Table 10 reports the continuous-depth result. The first-month sample contains 11,327 comparable endpoint-pair days; on 7,752 of them, the best stable weak-leg depth is below one-tenth of WETH weak-leg depth, and stablecoins carry 2.4% of route counts in that range. Within a bridge event, a 1 pp increase in Q_{et} is associated with +0.64 pp higher stable route share. Stablecoins carry 53.0% of routes when their weak-leg depth is at least WETH depth, and 69.9% when it is at least twice WETH depth. Deposited depth is therefore the proximate liquidity-formation margin connecting support to route allocation. The design identifies a relationship with local capital stocks; executable quotes and causal provider-supply shifts require separate measurement and design.

Panel C orders support and use in calendar time. Only 8.3% of the 865 support events use a stablecoin in the event's support set on the same day; 52.8% do within 30 days and 60.1% within 120 days. Among the 853 events with positive event-day WETH depth, first-month adoption is 42.6% when stable depth is below one-tenth of WETH depth and 84.1% once it reaches that threshold. The unadjusted difference is +41.5 pp, with standard error 4.9 pp. Conditioning on pre-event WETH route and value activity, endpoint direction, route scope, the supporting stablecoins, and event-month effects leaves a +25.2 pp difference, with standard error 5.8 pp. With two non-stablecoin endpoints, the difference remains +35.8 pp, with standard error 11.9 pp. Persistent support thus commonly precedes use of the supported stablecoin, while relative depth separates bridges that are followed by adoption from those that remain little used. The estimates are descriptive because route demand and provider capital are jointly determined.

Figure 4 follows the 246 events whose supported stablecoin is first used at least seven days after persistent support, so the entire pre-route week follows stable-bridge availability. Stablecoin bottleneck capital rises +0.86 log points over that week and changes −0.44 over the following week. Relative to the matched WETH bridge, the changes are +0.78 before route use and −0.46 after it. The pre-minus-post contrast is +1.24 and remains +1.29 after 5/95 winsorization. Capital formation is heterogeneous: 63.0% of events have a pre-route increase, the median is +0.07, and the ten largest positive changes contribute 41.3% of total positive accumulation. The path places route-specific capital before first observed use. Deposited-capital stocks do not isolate provider cash flow or intent, and the sample conditions on eventual adoption.

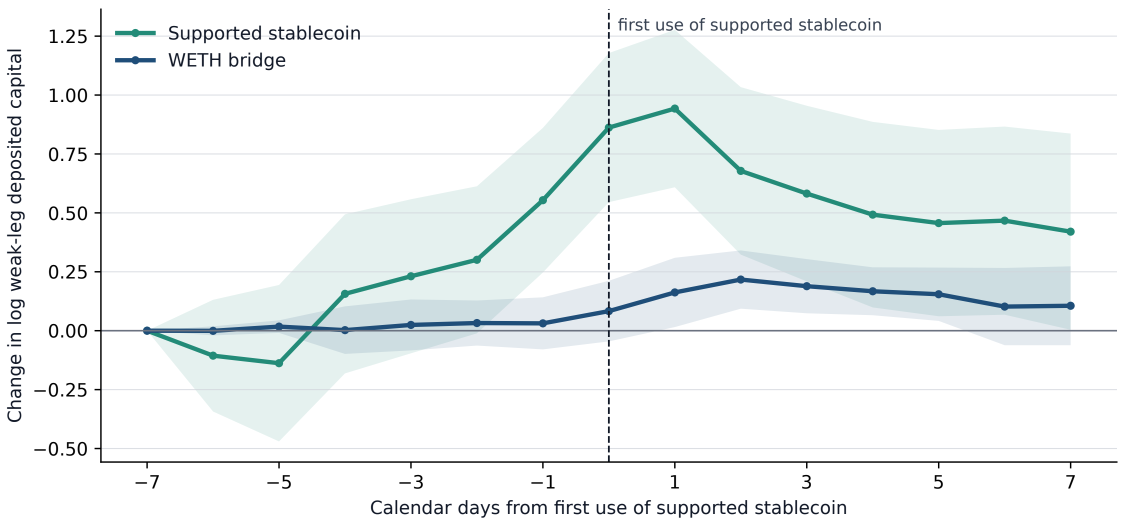
Table 11 separates pool activation from capital scaling over the pre-use week. The supported stablecoin's active-pool count grows by +0.05 log points more than the matched WETH leg, and the probability of at least one newly capital-positive pool is +6.9 pp higher. Pool activation remains the smaller adoption-day capital margin. Newly active pools hold 7.5% of supported-stablecoin weak-leg capital on the first-use day, leaving 92.5% in pools that were already capital-positive seven days earlier. Among events with continuing pools for both vehicles, supported-stablecoin capital in those pools grows by +0.16 log points more than WETH. Capital around first use therefore combines occasional pool activation with scaling inside existing pools.

Table 10: Stable-bridge establishment, relative depth, and route allocation

Outcome	First 30 days	Days 30–119
Panel A. Changes around first persistent stable support		
Stable route share [pp]	+7.93*** (1.88)	+8.68*** (2.57)
Stable supported-value share [pp]	+3.11** (1.53)	+6.94 (4.53)
$\log(1 +$ native routes per active day$)$	−0.20*** (0.04)	−0.39*** (0.06)
$\log(1 +$ total routes per active day$)$	−0.10** (0.04)	−0.29*** (0.05)
$\log(1 +$ native supported value per active day$)$	−0.17 (0.11)	−0.61*** (0.19)
$\log(1 +$ total supported value per active day$)$	+0.06 (0.10)	−0.32* (0.18)
Bridge events	865	818
Panel B. Stable-bridge competitiveness relative to WETH		
Relative-depth slope [pp per +1 pp]	+0.64*** (0.07)	+0.75*** (0.05)
Stable route share, depth $< 0.1\times$ WETH [%]	2.4	2.4
Stable route share, depth $\geq$ WETH [%]	53.0 (4.6)	61.2 (13.2)
Stable route share, depth $\geq 2\times$ WETH [%]	69.9 (2.3)	70.7 (9.3)
Active ordered-ultimate-pair days	11,327	17,027
Panel C. Adoption of supported stablecoin after persistent support		
Outcome	First 30 days	First 120 days
Supported stablecoin used, depth $< 0.1\times$ WETH [%]	42.6	50.5
Supported stablecoin used, depth $\geq 0.1\times$ WETH [%]	84.1	89.7
Difference [pp]	+41.55*** (4.95)	+39.17*** (4.48)
Difference with controls [pp]	+25.16*** (5.76)	+23.47*** (5.37)
Difference, no stablecoin endpoint [pp]	+35.76*** (11.89)	+40.26*** (11.66)

Note: The unit is an ordered endpoint pair and route scope around the first persistent DAI, USDC, or USDT bridge. Support requires positive prior-calendar deposited capital on both route legs in Uniswap V2 or SushiSwap V2 on at least 24 of the next 30 calendar days. Every endpoint pair has prior WETH-mediated routes and no earlier observed stablecoin route. In Panel A, the pre-period is the 30 calendar days before support; share rows report paired pp changes using harmonic pre/post route or supported-value mass, and intensity rows report paired log changes with equal event weights. Panel B compares the best stablecoin weak-leg capital with WETH weak-leg capital on post-event active days. The relative-depth slope is Equation (10); threshold rows are route-count-weighted means. Panel C records whether a route uses a stablecoin in the event's exact support set within 30 or 120 calendar days. Its stablecoin depth is restricted to that same support set and compared with the WETH bottleneck; 12 events without a positive WETH denominator are excluded. Panel C differences give every event equal weight. Controlled differences include log pre-event WETH route count and supported value per active day, along with event-month, route-scope, endpoint-direction, and supporting-stablecoin-set effects. The final row retains only events with two non-stablecoin endpoints. Parentheses contain standard errors two-way clustered by ordered endpoint pair and date. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital measures the prior-calendar pool stock and supplies a proxy for liquidity formation; executable quotes and causal provider-supply shifts require separate measurement and design.

Figure 4: Bridge capital accumulates before first use of the supported stablecoin



Note: The sample contains 246 ordered endpoint-pair and route-scope events whose first use of a stablecoin in the event's exact support set occurs 7 to 119 calendar days after persistent support. Each line is the equal-event mean change from day -7 in log(1 + deposited capital in USD) on the weaker route leg. Stablecoin depth is the largest bottleneck among the stablecoins in that support set; WETH depth is the matched incumbent bottleneck. Deposited capital is lagged to the prior calendar day. Shading gives 95% confidence intervals from standard errors two-way clustered by ordered endpoint pair and adoption date. The stable-minus-WETH pre-minus-post contrast is +1.24 with standard error 0.24; its 5/95 winsorized counterpart is +1.29 with standard error 0.18. The sample is selected on persistent support and first route use, so the paths order capital and use without identifying provider intent.

Table 11: Pool activation and capital scaling before stablecoin route use

Outcome	Stablecoin	WETH	Difference
Weak-leg deposited capital [log change]	$+0.76^{***}$	$+0.06$	$+0.70^{***}$
	(0.16)	(0.07)	(0.17)
Active pool count [log change]	$+0.04^{***}$	-0.01^{*}	$+0.05^{***}$
	(0.01)	(0.01)	(0.01)
Any newly active pool [pp]	$+7.32^{***}$	$+0.41$	$+6.91^{***}$
	(1.66)	(0.41)	(1.72)
New-pool share of adoption capital [pp]	$+7.47^{***}$	$+0.41$	$+7.14^{***}$
	(1.70)	(0.41)	(1.77)
Continuing-pool capital [log change]	$+0.23^{***}$	$+0.07^{*}$	$+0.16^{**}$
	(0.07)	(0.04)	(0.08)

Note: The sample is the 246 bridge events in Figure 4. For each event, the supported stablecoin is the member of the event's exact support set carrying the most routes on its first-use day and is held fixed from day -7 through day zero. Stablecoin and WETH outcomes use each vehicle's weaker adoption-day route leg. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; a continuing pool has positive capital on both dates. This definition records pool-level capital activation and does not imply that the pool contract was newly deployed. Each cell is an equal-event mean. The Difference column pairs the supported stablecoin with WETH within the event. Standard errors in parentheses are two-way clustered by ordered endpoint pair and adoption date. Difference-column significance for the four pool-composition rows uses Holm-adjusted p-values across those rows; the deposited-capital row and the Stablecoin and WETH columns use unadjusted p-values. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. The capital, pool-count, and pool-activation rows contain 246 observations per vehicle and difference. The new-pool capital-share row contains 241 stablecoin, 243 WETH, and 238 paired observations; the continuing-pool row contains 223, 242, and 219. Deposited-capital stocks do not identify provider cash flow or intent.

The local bridge-depth analysis supports that earlier-margin interpretation. Let $c = (i, j, t, s)$ denote an ordered endpoint-pair, date, and route-scope opportunity; let v index vehicles; and define $B_{vc} = \log(1 + \min\{K_{iv,t-1}, K_{vj,t-1}\})$, the prior-calendar deposited capital on the weaker of the two route legs. The main route-choice specification is

$$\begin{aligned} R_{vc} = \beta_B B_{vc} + \theta_0 \log(1 + N_{v,t}^{(-c)}) + \theta_1 \log(1 + N_{v,t-30}) \\ + \theta_2 \log(1 + P_{v,t-30}) + \alpha_c + \delta_v + \varepsilon_{vc}, \end{aligned} \tag{11}$$

where R_{vc} is the vehicle's share of five-vehicle route mass, $N_{v,t}^{(-c)}$ is same-day leave-one-opportunity-out route reach, and $N_{v,t-30}$ and $P_{v,t-30}$ are prior-30-day route and endpoint-pair reach. The regression weights observations by five-vehicle route count, absorbs opportunity and vehicle effects, and clusters by ordered endpoint pair and date. For each opportunity in which at least two of the five headline vehicles have both atomic legs supported, we rank vehicles by B_{vc}. The deepest supported bridge carries 84.1% of five-vehicle route mass, rising from 75.5% in 2024 to 86.5% in 2026, across 58,483 opportunity cells and 2,923 ordered endpoint pairs. With opportunity and asset effects, a one-log-point larger weaker-leg capital stock is associated with +6.67 pp higher route share, with standard error 1.21 pp; the corresponding stablecoin total slope is +7.73 pp, with standard error 1.46 pp. Adding the reach controls in equation (11) leaves the local-depth coefficient at +6.79 pp, with standard error 1.15 pp; same-day global route reach contributes +9.10 pp, with standard error 2.81 pp. Restricting the same vehicle set to first-observed endpoint pair dates, and retaining zero-depth alternatives as part of the birth choice set, gives the entry margin directly: across 835 vehicle rows in 167 entry opportunities, a one-log-point larger local bridge predicts +5.35 pp higher route share and +5.28 pp higher selection probability, with standard errors 0.51 pp and 0.57 pp. Prior-30-day route reach is estimated at +8.78 pp with standard error 5.72 pp, so local depth adds information beyond existing vehicle popularity. A two-leg bottleneck specification gives the physical-market version of the mechanism: holding the stronger leg and reach fixed leaves the weaker-leg slope at +6.61 pp, with standard error 1.17 pp, while a one-log-point leg imbalance predicts −5.43 pp route share, with standard error 1.14 pp. The local-depth slope also survives all 5 leave-one-vehicle reruns of the same horse race, with the weakest retained slope +5.98 pp and standard error 1.54 pp, which makes the result robust to every single-vehicle omission. Inside the stable-only supported race, 12,048 opportunity cells across 296 ordered endpoint pairs compare DAI, USDC, and USDT directly. With the same opportunity effects, local depth, and reach controls, the 2026 USDC interaction is +34.65 pp with standard error 17.18 pp, the USDT interaction is +55.29 pp with standard error 13.17 pp, and local depth remains +6.01 pp with standard error 1.97 pp. Conditional on all three stablecoin bridges being feasible in the same opportunity, the 2026 allocation is concentrated in USDC and USDT relative to DAI. Deposited capital remains a predictive state variable. Route choice is therefore organised around both vehicle identity and endpoint-pair-specific two-leg depth where endpoint-pair entry and activity growth create scope for vehicle formation.

Table 12: Dynamic feedback between route use and local bridge depth

Feedback equation	30 days	120 days
$R_{b,t} \to \Delta B_{b,t+h}$, pooled [log points]	+0.020*** (0.005)	+0.097*** (0.013)
$R_{b,t} \to \Delta B_{b,t+h}$, stable [log points]	+0.024*** (0.005)	+0.115*** (0.012)
$B_{b,t} \to \Delta R_{b,t+h}$, pooled [pp]	+1.262*** (0.189)	+1.595*** (0.271)
$B_{b,t} \to \Delta R_{b,t+h}$, stable [pp]	+1.773*** (0.228)	+2.463*** (0.292)
Rows / ultimate pairs / dates	51,240 / 503 / 303	18,304 / 340 / 123
Asset and date effects		Yes
Two-way clustered covariance		Ordered ultimate pair and date

Note: Each cell reports a predictive coefficient with two-way clustered standard errors in parentheses. For source–vehicle–destination bridge b, $R_{b,t}$ is the vehicle share of five-vehicle route mass and $B_{b,t} = \log(1 + \min\{K_{iv,t}, K_{vj,t}\})$ is deposited capital on the weaker route leg; Δ is the exact-horizon value minus the origin-date value. Route-use rows scale $R_{b,t}$ by 10 pp and report $\Delta B_{b,t+h}$ in log points. Bridge-depth rows scale $B_{b,t}$ by one log point and report $\Delta R_{b,t+h}$ in pp. All regressions absorb vehicle and origin-date effects, control for the other current state variable, weight by five-vehicle route count, and cluster by ordered endpoint pair and date. Stable rows add the stablecoin interaction and report the stable total. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. These local feedback regressions use observational variation.

The USDC and USDT interactions answer a different conditional question: among stablecoin bridges already supported in the same opportunity, which issuer carries more route flow in 2026? They rank feasible stablecoin alternatives after availability and do not date bridge establishment.

A dynamic bridge analysis gives the feedback version of the volume–cost mechanism in Krugman (1980). Let b index a source–vehicle–destination bridge, $R_{b,t}$ be the vehicle's share of five-vehicle route mass in that local opportunity, and $B_{b,t}$ be the log weak-leg bridge depth. The two predictive equations are

$$\begin{aligned} \Delta B_{b,t+h} &= \beta_R R_{b,t} + \gamma_R B_{b,t} + \alpha_c + \delta_t + \varepsilon_{b,t+h}, \\ \Delta R_{b,t+h} &= \beta_B B_{b,t} + \gamma_B R_{b,t} + \alpha_c + \delta_t + \eta_{b,t+h}. \end{aligned} \tag{12}$$

Matching the same bridge at exact future horizons, a 10 pp higher current route share predicts +0.020 log points more bridge-depth growth after 30 days and +0.097 after 120 days. In the other direction, a one-log-point larger current bridge predicts +1.26 pp more route-share growth after 30 days and +1.60 pp after 120 days. The feedback margin is stronger for stable vehicles: the corresponding 120-day stable route-share to depth-growth coefficient is +0.115 for a 10 pp route-share difference, and the stable depth to route-share coefficient is +2.46 pp. These rows absorb vehicle and origin-date effects and cluster by ordered endpoint pair and date. Consistent with constrained and active liquidity provision (Lehar and Parlour, 2025; Huang et al., 2025), the estimates identify predictive comovement: route use and local bridge depth forecast one another inside the same bridge. One-for-one provider funding lies outside the design.

Table 12 reports the coefficients from Equation (12).

The aggregate allocation of capital does not move in lockstep with vehicle use. In the daily five-vehicle V2 panel, WETH is the largest deposited-capital vehicle on 100.0% of 2,239 days, while stablecoins lead excess use on 81.3% days. From 2024 to 2026, stablecoins' share of deposited capital falls from 17.6% to 11.1%, while their share of intermediary route counts rises from 18.1% to 46.6%. Their 2026 route share is 4.2× their capital share. Holding the date fixed and controlling for endpoint share, venue count, and pool count leaves stablecoins with +13.3 pp more route share than capital share.

To examine how providers adjust after such a mismatch, let $G^S_{c,t}$ denote stable vehicle c's route share minus its deposited-capital share on date t. For a provider outcome observed h days later, we estimate

$$Y_{c,t+h} = \beta G^S_{c,t} + X'_{c,t}\Gamma + \alpha_c + \delta_t + \varepsilon_{c,t+h}, \tag{13}$$

where $X_{c,t}$ contains the origin-day controls named in Table 13. The table reports the association for a 10 percentage point larger gap. The vehicle and date effects compare stable vehicles at different points in their own route-capital imbalance while absorbing common market conditions.

The V2 rows locate capital adjustment in the vehicle's pool network. A 10 pp shortfall predicts +1.65 pp higher capital share and +0.23 positions of capital-rank catch-up after 120 days. The same-pool estimate is +0.72%, while capital in pools already active on the origin date rises +12.39%. Capital therefore converges mainly through reallocation across incumbent pools, with little evidence of immediate funding in the particular pool that generated the route shortfall. Origin capital share enters both the gap and the future change, so these stock coefficients include mechanical mean reversion.

The V3 rows identify activity on both sides of the provider's position. A 10 pp shortfall predicts +67.89% more mint events and +64.20% more burn events over 30 days, but only +1.82 pp in the net mint-minus-burn balance. After controlling for the vehicle's current action and provider-day counts, future provider-day activity rises +6.02% over 120 days. Fee yield changes by −0.27%. The response is therefore provider turnover and repositioning, with no corresponding increase in fee yield.

The USDC stress of March 2023 shows why route demand and provider capital must be separated. Stablecoin route share rises by +11.4 pp inside the five-vehicle set, while stablecoin deposited-capital share moves −2.0 pp during the same window and −3.3 pp afterward. The episode combines peg stress, arbitrage, and demand, so it does not isolate a supply shock. It does show that an immediate route-demand increase can coincide with capital moving away from stablecoins.

V4 exposes an additional adjustment margin because the singleton records the complete swap path while settling only final net balances. A 10 pp stable route-capital shortfall is associated with +5.40 pp more multi-leg transaction share, +4.50 pp more same-asset internal-leg share, and +2.32 pp more gross-to-net reduction on the same day. Conditional on current vehicle-token-side flow, the shortfall predicts +30.91% more future LP flow after 120 days; add and remove flow rise by +23.99% and +40.07%. Conditional on current activity, LP actions and sender-days rise by +30.04% and +12.64%. These estimates connect visible vehicle demand to provider flow and position management inside V4. They remain predictive because route demand, pool profitability, and provider expectations can move together.

Within V4, the singleton's accounting intensity permits a more direct comparison of trading

Table 13: Liquidity-provision outcomes and stable route-capital gaps

Margin	Outcome ($Y_{c,t+h}$)	Horizon	Effect of +10 pp in $G^S_{c,t}$	Obs.	Clusters
V2 stock	Capital share change [pp]	120d	$+1.4^{***}$ (0.2)	10,595	2,119
V2 stock	Log deposited capital [log points]	120d	$+0.205^{***}$ (0.031)	10,595	2,119
Portfolio rank	Capital-rank catch-up [rank positions]	120d	$+0.225^{***}$ (0.036)	10,595	2,119
Same pool	Same-pool capital [log points]	120d	$+0.007$ (0.009)	174,712	2,113
Pool footprint	Incumbent-pool capital [log points]	120d	$+0.124^{**}$ (0.054)	10,417	2,117
Rent incidence	V3 fee yield [log points]	120d	-0.003 (0.005)	208,360	189
V3 churn	Mint events [log points]	30d	$+0.679^{***}$ (0.037)	11,195	2,239
V3 churn	Burn events [log points]	30d	$+0.642^{***}$ (0.035)	11,195	2,239
V3 churn	Net mint-burn balance [events]	30d	$+0.018^{***}$ (0.002)	11,195	2,239
V3 activity-controlled	Provider-day activity [log points]	120d	$+0.060^{***}$ (0.010)	11,195	2,239
V4 accounting	Multi-leg transaction share [pp]	same day	$+5.4^{***}$ (0.6)	2,599	522
V4 accounting	Internal same-asset share [pp]	same day	$+4.5^{***}$ (0.4)	2,599	522
V4 accounting	Gross-to-net reduction share [pp]	same day	$+2.3^{***}$ (0.4)	2,599	522
V4 LP flow	Gross vehicle-side flow [log points]	120d	$+0.309^{***}$ (0.041)	2,592	521
V4 LP flow	Add flow [log points]	120d	$+0.240^{***}$ (0.036)	2,592	521
V4 LP flow	Remove flow [log points]	120d	$+0.401^{***}$ (0.047)	2,592	521
V4 activity-controlled	LP actions [log points]	120d	$+0.300^{***}$ (0.024)	2,599	522
V4 activity-controlled	Sender-days [log points]	120d	$+0.126^{***}$ (0.016)	2,599	522

Note: Each cell reports β from Equation (13), the association for a 10 pp larger stablecoin route-minus-capital gap $G^S_{c,t}$, with clustered standard errors in parentheses. Capital-share change is target-date minus origin-date share within the five-vehicle V2 capital total. Because origin capital share enters both this outcome and $G^S_{c,t}$, capital-share and capital-rank rows retain a mechanical mean-reversion component and describe convergence; identifying provider reallocation requires exogenous variation. Log-point outcomes are changes in $\log(1+q)$, where q is deposited capital in dollars, event count, provider-day count, sender-day count, vehicle-token-side LP flow in dollars, or fee yield as named by the row. Capital-rank catch-up is origin rank minus future rank, so a positive value is an improvement; net mint-burn balance is the future mint-event count minus burn-event count. V4 accounting outcomes are same-day transaction shares in pp. Vehicle-day rows absorb vehicle and origin-date effects unless the row label implies a pool-level design: same-pool capital absorbs pool–vehicle and origin-date effects, V3 fee yield absorbs pool and origin-date effects, and V4 rows use V4-active origin vehicle-days. V3 activity-controlled and V4 activity-controlled rows include current origin-day action, origin-count, and sender-count controls; V4 flow rows include current vehicle-token-side LP-flow controls. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. These predictive mechanism regressions use observational variation.

Table 14: V4 flash accounting and subsequent liquidity provision

Flash-accounting proxy ($M_{c,t}$)	Future LP flow [log points]	Vehicle-linked TVL [log points]	LP actions [log points]	Flow narrow/medium [pp]	Flow broad [pp]	Action narrow/medium [pp]	Action wide/very-wide [pp]
Internal same-asset share	$+0.035^{*}$ (0.020)	$+0.265^{***}$ (0.031)	$+0.071^{***}$ (0.013)	$+1.398^{***}$ (0.137)	-1.398^{***} (0.137)	-2.460^{***} (0.384)	$+2.889^{***}$ (0.354)
Multi-leg transaction share	$+0.076^{**}$ (0.032)	$+0.123^{***}$ (0.018)	$+0.058^{***}$ (0.017)	$+1.165^{***}$ (0.104)	-1.165^{***} (0.104)	-1.867^{***} (0.203)	$+1.808^{***}$ (0.224)
Gross-to-net reduction share	$+0.139^{***}$ (0.041)	+0.026 (0.031)	$+0.044^{**}$ (0.019)	$+1.223^{***}$ (0.200)	-1.223^{***} (0.200)	+0.362 (0.373)	+0.021 (0.328)
Observations / date clusters							2,015 / 403
Asset and date effects							Yes
Origin-day activity controls							Yes

Note: Each cell reports the effect of a 10 pp increase in flash-accounting proxy $M_{c,t}$ on the column outcome, with standard errors in parentheses. Multi-leg share is the fraction of vehicle-touching transactions with at least two swap legs; internal same-asset share is the fraction in which the vehicle asset appears on at least two legs; gross-to-net reduction is $1 - |\sum_\ell \Delta q_{c\ell}|/\sum_\ell |\Delta q_{c\ell}|$, aggregated from vehicle-token amount changes within each singleton transaction. The unit is a vehicle-day in Uniswap v4. All columns use the 120-day horizon, absorb vehicle and origin-date fixed effects, include origin-day swap activity, current vehicle-token side LP flow, current vehicle-linked TVL, current LP actions, and current range-composition controls, and cluster by origin date. LP flow is the admissible dollar value of vehicle-token sides in modify-liquidity events. Vehicle-linked TVL is full-pool provider TVL linked to a vehicle token side after address-resolution and physical TVL/volume bounds. Flow-share columns partition future dollar-valued modify-liquidity flow into narrow-or-medium versus broad ranges; action-share columns partition future modify-liquidity events into narrow-or-medium and wide-or-very-wide placements. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

paths and later provider behavior:

$$Y_{c,t+h} = \beta M_{c,t} + X'_{c,t}\Gamma + \alpha_c + \delta_t + \varepsilon_{c,t+h}. \tag{14}$$

In Equation (14), $M_{c,t}$ is internal same-asset routing, multi-leg transaction share, or gross-to-net settlement reduction. The controls contain origin-day swap activity, vehicle-token-side LP flow, vehicle-linked reported liquidity, LP actions, and range composition. Vehicle and date effects compare vehicle assets on the same day. Table 14 uses the 120-day horizon, where the earlier provider responses are most visible.

Internal same-asset routing and multi-leg transaction share forecast more LP flow, reported liquidity, and LP actions. Dollar value shifts toward narrow or medium ranges, while the number of position updates shifts toward wider ranges. Gross-to-net reduction predicts more LP flow but little change in reported liquidity. These different signs separate two provider margins: concentrated placement of value and broader position-management activity.

Table 15 asks whether these margins are stronger when stablecoin route demand is high relative to deposited capital. The first two rows associate larger shortfalls with more reported liquidity and LP actions, together with a shift in position updates toward wider ranges. Gross-to-net reduction reverses those signs. The accounting proxies therefore carry different economic content even though each is enabled by singleton settlement. The corresponding flow interactions in Appendix G show

Table 15: Stable shortfalls, V4 flash accounting, and LP repositioning

Interaction term ($G^S_{c,t} \times M_{c,t}$)	Vehicle-linked TVL [log points]	LP actions [log points]	Narrow/medium ranges [pp]	Wide/very-wide ranges [pp]
Stable gap × internal same-asset share	$+0.292^{***}$ (0.079)	$+0.082^{***}$ (0.027)	-6.531^{***} (0.720)	$+6.189^{***}$ (0.657)
Stable gap × multi-leg transaction share	$+0.225^{***}$ (0.035)	$+0.132^{***}$ (0.013)	-4.539^{***} (0.442)	$+4.438^{***}$ (0.381)
Stable gap × gross-to-net reduction share	-0.667^{***} (0.105)	-0.121^{***} (0.043)	$+4.411^{***}$ (1.261)	-4.405^{***} (1.126)
Observations / date clusters				1,209 / 403
Asset and date effects				Yes
Origin-day activity controls				Yes

Note: Each cell reports $G^S_{c,t} \times M_{c,t}$, the interaction of a 10 pp larger stable route-minus-capital gap with a 10 pp larger V4 accounting proxy; $G^S_{c,t}$ is stablecoin route share minus deposited-capital share and the three $M_{c,t}$ proxies are defined in Table 14. Log outcomes are in log points; range outcomes are in pp. The unit is a stable vehicle-day in Uniswap v4. All columns use the 120-day horizon, absorb vehicle and origin-date fixed effects, include origin-day swap activity, current vehicle-token side LP flow, current vehicle-linked TVL, current LP actions, and current range-composition controls, and cluster by origin date. Vehicle-linked TVL is full-pool provider TVL linked to a vehicle token side after address-resolution and physical TVL/volume bounds. Range shares partition future modify-liquidity events into narrow-or-medium, wide-or-very-wide, and full-range placements. The estimates are predictive associations from observational variation. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

less gross and add-side flow when internal or multi-leg routing is high, and more removal turnover when net settlement compression is high.

The protocol comparison uses the same vehicle and day, which removes vehicle-date demand from the V3-versus-V4 contrast. It restricts the sample to dates with observed V4 singleton swap activity and stacks the V3 mint/burn outcome with the corresponding V4 modify-liquidity outcome. For protocol $p \in \{V3, V4\}$, vehicle v, date t, and horizon h, we estimate

$$\begin{aligned} Y_{pvt+h} = \alpha_{v,t} + \delta_p + \beta_h \, \mathbf{1}\{p = V4\} \, \mathbf{1}\{v \in \mathcal{S}\} \, G_{v,t} \\ + \boldsymbol{\gamma}_h' \mathbf{X}_{pvt} + \varepsilon_{pvt+h}, \end{aligned} \tag{15}$$

where $G_{v,t}$ is route share minus deposited-capital share and $\mathcal{S}$ is the stablecoin set. Vehicle-date effects compare the two protocols for the same vehicle and date; protocol effects absorb level differences. The coefficient β_h is the V4-minus-V3 difference in the association between a stablecoin shortfall and future provider behavior.

Table 16 shows that the response is strongest on margins that require active position management. A 10 pp stablecoin shortfall predicts 0.320 log points more V4-relative-to-V3 LP actions and 0.115 log points more active provider origins after 120 days. Gross and add-side vehicle-token flow differentials are small, while removal flow rises by 0.046 log points and narrow-or-medium flow share rises by 1.126 pp after 30 days. Reported liquidity grows by 0.637 log points and the pool footprint by 0.275 log points at 120 days. Thus the V4 difference is not a uniform increase in deposits. It combines more provider participation, delayed removal turnover, short-run concentration of value, and expansion across pools. This pattern is consistent with providers reallocating positions when vehicle demand exceeds committed capital, as in the adjustment-cost and inventory mechanisms

Table 16: Stablecoin shortfalls and liquidity provision in V4 relative to V3

Provider outcome	Horizon	V4-minus-V3 effect	Obs. / dates
LP actions [log points]	120 days	$+0.320^{***}$ (0.013)	5,198 / 522
Active provider origins [log points]	120 days	$+0.115^{***}$ (0.010)	5,198 / 522
Gross vehicle-side flow [log points]	120 days	+0.029 (0.022)	5,188 / 521
Add-side flow [log points]	120 days	+0.012 (0.023)	5,188 / 521
Remove-side flow [log points]	120 days	$+0.046^{**}$ (0.022)	5,188 / 521
Narrow/medium flow share [pp]	30 days	$+1.126^{***}$ (0.133)	5,188 / 521
Reported liquidity [log points]	120 days	$+0.637^{***}$ (0.042)	1,856 / 188
Pool footprint [log points]	120 days	$+0.275^{***}$ (0.022)	1,856 / 188
Vehicle-date and protocol effects			Yes
Origin provider controls			Yes

Note: Each row reports β_h from Equation (15), the V4-minus-V3 difference associated with a 10 pp larger stablecoin route-minus-capital gap, with standard errors in parentheses. The stacked unit is a protocol–vehicle–date row restricted to dates with observed V4 singleton swap activity. All specifications absorb vehicle-date and protocol fixed effects, include the corresponding protocol's origin-day provider controls, and cluster by date. V3 flows value mint and burn token sides; V4 flows value modify-liquidity token sides. Reported liquidity is full-pool TVL linked to the vehicle token and can count a vehicle–vehicle pool in both vehicle series. The estimates are conditional protocol differences and do not identify the effect of V4 adoption. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively. Appendix G reports every outcome and horizon.

of Lehar and Parlour (2025); Huang et al. (2025). The reported-liquidity measure assigns full-pool TVL to each vehicle token. The design identifies conditional reorganization; a causal adoption effect requires exogenous protocol timing.

5.5 Protocol architecture

Protocol design changes the set of routes that can be formed. Uniswap v1 created pools between an ERC-20 token and ETH, so token-to-token trades crossed ETH; Uniswap v2 allowed arbitrary ERC-20–ERC-20 pools and made direct token-to-token trading possible.[6] Forced intermediation was material while the constraint lasted: token-to-token trades routed through ETH account for 8.0% of Uniswap v1 swap transactions and 8.1% of its ETH volume over the venue's life, and roughly one trade in ten in 2019 and 2020, when the venue was most active. A transaction from the crash weekend of March 2020 shows the mandate operating: with no direct pool between MKR and DAI, a trader selling 250 MKR received 439.13 ETH at one exchange contract and spent the identical ETH amount at a second for 49,892.40 DAI.[7] Native-asset pairing remained pervasive after this mandate ended. The share of single-leg Uniswap v2 trades executed in WETH pools was 95.1% in 2020 and 95.5% in 2026, remaining between 93.4% and 97.9% in every intervening year. Among the 477,633 token pairs that ever traded on Uniswap v2, 97.1% include WETH, and WETH's share of newly traded token pairs rose from 84.1% in 2020 to 97.9% in 2026. This persistence is consistent with deep WETH pools, launch conventions, and inherited search paths. Separating those forces from the original protocol constraint requires comparison among token pairs that had the same alternatives available.

Uniswap v4 separates route choice from physical settlement more sharply. All pools reside in one singleton, and flash accounting records balance changes internally before settling the final net amounts at the end of a call.[8] An economic route can therefore cross an intermediary pool without generating an external token transfer at every step. Decoded pool actions preserve the trading path despite this netting, while the architecture changes how liquidity, fees, and settlement are organised around that path.

The route evidence leaves an explanation six features to account for. The stablecoin gain occurs mainly across endpoint pairs, with little repeated substitution within matched endpoint pairs; it appears on short and long paths alike and on routes that stay within one exchange as well as routes that do not; the incidence of intermediation remained flat when the software that searches for paths was rewritten, though the exchange set it reached did widen; the gain is present where every leg was priced by the same unchanged constant-product rule, and absent from the venue whose invariant is built for pegged assets, which carries no intermediation at all; local two-leg depth predicts which vehicle carries route mass inside an opportunity; and the measured WETH-pairing persistence

[6] See Adams, Zinsmeister, and Robinson, *Uniswap v2 Core*, pp. 1–2.

[7] Transaction `0x4dca160d…96a0ca16`, block 9,674,728, March 15, 2020. The trace is selected by a deterministic rule, the largest ETH amount routed among the 129,810 mandate-era token-to-token transactions whose two ETH legs match exactly, and the token identities are verified against the public transfer record.

[8] See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

shows that route opportunities can outlast the protocol rule that first created them. Together they narrow the set of viable explanations. A viable explanation must instead account for why newly active and growing endpoint pairs attach themselves to stablecoins.

6 Conclusion

Stablecoins become dominant vehicle currencies on decentralised exchanges through pair formation and local liquidity. Between matched January–June periods in 2024 and 2026, their share among native and stable intermediaries rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by routed value. Positive and negative substitutions offset inside continuing endpoint pairs. Activity reallocation toward stablecoin-intensive pairs and the entry and exit of pairs account for the aggregate change.

The vehicle used at pair entry predicts later use. A 1 pp higher initial stablecoin share predicts +0.74 pp higher stablecoin share over 120 days, and the relation remains when the sample excludes WETH endpoints, contains no direct route at entry, or weights routes by supported value. Capital typically precedes first use of the supported stablecoin after persistent bridge support appears. First-month adoption is 42.6% when stablecoin depth is below one-tenth of WETH depth and 84.1% once it reaches that threshold. The gap survives controls for prior WETH activity and remains among pairs with two non-stablecoin endpoints. Once stable and native bridges coexist, deposited capital on the weaker leg predicts route choice. At first use, 92.5% of the supported-stablecoin bottleneck capital lies in pools active one week earlier, and continuing-pool capital has grown more than in the matched WETH bridge. These findings locate persistence in the trading relationships and liquidity pools that form around a vehicle; they do not require widespread switching away from the incumbent inside established pairs.

This distinction extends beyond decentralised exchange. Cross-border payment systems still require an institution to exchange the source currency into the destination currency. Project Nexus assigns this function to an FX provider, and Project Rialto uses a vehicle currency when direct exchange is unavailable (Bank for International Settlements Innovation Hub, 2024, 2025). The evidence here shows why aggregate dominance in such systems can change before incumbent relationships switch: new relationships can inherit a different liquid bridge. Models and empirical tests that hold the trading pair fixed describe vehicle substitution, while aggregate currency shares also reflect which pairs enter, grow, and receive liquidity.

A vehicle currency is made at two connected points: when a new trading relationship adopts it and when liquidity providers make the required legs competitive. Aggregate dominance records both.

A From contract events to economic routes

Route construction begins with the contract and event format of each exchange. We identify pools from the protocol's factory or registry, map contract addresses to tokens, and convert raw integer amounts using the token precision that applied at the time of the trade. Contract addresses define token identity because unrelated tokens can share a symbol and the same economic asset can appear in native and wrapped form. ETH and WETH are consolidated only after this address-level reconstruction; every other token remains distinct unless a stated economic classification combines it. Within each transaction, pool-swap legs are ordered by block, transaction, and log position. The resulting token flows reveal the exchange that occurred. A single connected chain gives one route and therefore one endpoint exchange between its endpoints; parallel chains identify an order split, and disconnected event groups remain separate. Because one transaction can contain unrelated calls as well as trades across several exchanges, this procedure also prevents two simultaneous routes from being mistaken for one longer route. The output records the sequence of token pairs and pools traversed, together with the endpoint pair. It does not infer that sequence from the transaction sender or router label.

A complete token sequence is sufficient for the route-count sample. Value weighting additionally requires coherent amounts on every leg and a common dollar valuation. A failed valuation removes the observation from the value calculation while preserving the observed route. Appendix F gives the separate exclusion for self-returning paths.

Historical token precision cannot be verified for 13 tokens traded in 22 constant-product pools. Reserve-reconstruction analyses omit those pools. An upper-bound check removes every route involving any of the 13 tokens: the resulting bound is 5,950 routes and \$10.91 million of route value. Only 1,036 of those routes use one of the affected tokens as an intermediary, representing \$1.284 million, or 0.00043% of value on routes whose legs can all be valued. All 13 tokens lie outside the native and stablecoin groups used in the main comparison. The exclusion affects exact reserve reconstruction; the route-composition result is unchanged.

B Pricing models and validation

The counterfactual comparison requires an exact-input quote from each pool's state and the proposed input amount. Because pool families use different invariants and arithmetic, we validate their pricing models separately. For selected executed swaps, we reconstruct the state immediately beforehand and compare predicted with realised output. Curve's amplification coefficient and Balancer's weight ratio are unavailable in the historical subgraph records, so we estimate them on one set of trades and evaluate the resulting quotes on held-out trades. A route enters the cost comparison only when the relevant pool states and pricing models are available before the transaction.

B.1 Constant-product pools

Uniswap v2 and SushiSwap v2 price the product of reserves (Lehar and Parlour, 2025). Let R_{i,p,e^-} and R_{o,p,e^-} denote pool p's input- and output-token reserves immediately before transaction e, Δ_i the token input, f_p the fee rate, and $\varphi_p = 1 - f_p$ the fraction of input retained for pricing. An exact-input swap moves the pool along

$$(R_{i,p,e^-} + \varphi_p \Delta_i)(R_{o,p,e^-} - \Delta_o) = R_{i,p,e^-} R_{o,p,e^-},$$

which inverts in closed form to

$$\Delta_o = \frac{\varphi_p \Delta_i R_{o,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i}, \qquad \varphi_p = \tfrac{997}{1000},$$

the 30 basis point fee both venues charge, applied on chain as the integer ratio above.[9]

A multi-leg route composes these one hop at a time, threading the output token of one pool into the next. The cost comparison evaluates the dollar outputs of the best direct and intermediated routes at the same state and dollar input notional,

$$10^4 \Delta C^D_{i,o,k,q,e} = 10^4 \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}}.$$

Both outputs must be computed from one reconstructed pre-transaction state. A realised trade supplies the output of the chosen route alone; the direct and intermediary alternatives require reconstructed states for every pool in the comparison. Quoting each executed constant-product swap against the state immediately before it gives a median absolute error of 0.0000% across 8,024 swaps; 96.7% of quotes are within 1% and 95.2% are within 0.01% of realised output.

B.2 Concentrated liquidity

Uniswap v3 distributes liquidity across price ranges, so executable depth depends on the active range and on the initialized ticks reached by a trade (Adams et al., 2021). Within one active range, let $\widetilde{R}_{i,p,e^-}$ and $\widetilde{R}_{o,p,e^-}$ denote the virtual reserves of the input and output assets. Liquidity L_{p,e^-} and the output-per-input marginal price ρ_{p,e^-} satisfy

$$\widetilde{R}_{i,p,e^-} = \frac{L_{p,e^-}}{\sqrt{\rho_{p,e^-}}}, \qquad \widetilde{R}_{o,p,e^-} = L_{p,e^-} \sqrt{\rho_{p,e^-}}.$$

If f_p is the pool fee and $\varphi_p = 1 - f_p$, an exact input Δ_i returns

$$\Delta_o = \frac{\varphi_p \Delta_i \widetilde{R}_{o,p,e^-}}{\widetilde{R}_{i,p,e^-} + \varphi_p \Delta_i}$$

[9]The SushiSwap v2 pool contract implements the same adjustment; see `UniswapV2Pair.sol`.

Table 17: Concentrated-liquidity quoter, by direction and tick crossing

Validation group	Swaps	Median	90th pct.	Maximum	Within 1%
token0 in, no tick crossed	2,290	0	0	9.98×10^{-4}	100.0%
token0 in, tick crossed	57	0	0	3.45×10^{-7}	100.0%
token1 in, no tick crossed	1,816	0	1.69×10^{-8}	1.48×10^{-4}	100.0%
token1 in, tick crossed	55	0	5.44×10^{-9}	8.74×10^{-9}	100.0%
Pooled	4,218	0	1.15×10^{-14}	9.98×10^{-4}	100.0%

Note: Absolute quote error in percent of realised output. The sample contains 4,218 realised Uniswap v4 swaps on three dates across 13 pools and nine token pairs, at static fee tiers 0, 7, 8, 100, 500, 10,000 and 20,000. Each quote uses liquidity events ordered strictly before the swap and the price state left by the preceding swap in the same pool.

until the trade reaches the next initialized tick. Range boundaries lie on a geometric grid, and crossing tick j updates active liquidity by its signed change,

$$\sqrt{\rho(j)} = 1.0001^{j/2}, \qquad L_{p,e^-} \mapsto L_{p,e^-} \pm \Delta L_j,$$

with the sign determined by the direction of the trade. The calculation then continues from the new active range until the input is exhausted. The quoter uses the contract's integer Q96 arithmetic, rounding input amounts up and output amounts down.

The active tick map is accumulated from every mint and burn strictly before the validation day, after which same-day liquidity changes and swaps are replayed together by block and log index. Consecutive swaps in one pool supply the price state: the earlier event leaves `sqrtPriceX96` and `tick` immediately before the later swap, while the interleaved event stream supplies active liquidity. We report errors separately by input direction and tick crossing because those cases use different arithmetic branches. Table 17 applies this validation to Uniswap v4. V4 retains concentrated-liquidity pricing for static-fee pools but implements it through a singleton and flash accounting.[10] The validated static fee tiers include 0, 7, 8, and 20,000 hundredths of a basis point. Hook-bearing and dynamic-fee pools are excluded because the raw records do not identify their transaction-level cash flows and realised fees; Appendix E reports their share of v4 trading.

Every direction and tick-crossing row of Table 17 reproduces realised output within 0.001% in the Maximum column.

B.3 StableSwap pools

Curve interpolates between a constant-sum curve, which is flat and charges no slippage, and the constant-product curve of Section B.1.[11] For n tokens with normalized pre-transaction balances

[10] See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

[11] See Egorov, *StableSwap—Efficient Mechanism for Stablecoin Liquidity*.

R_{j,p,e^-}, amplification A_p, and invariant D_p,

$$A_p n^n \sum_j R_{j,p,e^-} + D_p = A_p D_p n^n + \frac{D_p^{n+1}}{n^n \prod_j R_{j,p,e^-}},$$

where $A_p \to 0$ recovers the constant product and $A_p \to \infty$ recovers the constant sum. The contract solves D_p by Newton iteration and then solves the same invariant for the post-trade output balance $R'_{o,p,e}$ after R_{i,p,e^-} is increased by Δ_i. Defining $D_\Pi = D_p \prod_j D_p/(nR_{j,p,e^-})$, the two loops are

$$D_p \leftarrow \frac{(A_p n \sum_j R_{j,p,e^-} + nD_\Pi)D_p}{(A_p n - 1)D_p + (n+1)D_\Pi}, \qquad R'_{o,p,e} \leftarrow \frac{(R'_{o,p,e})^2 + c}{2R'_{o,p,e} + b - D_p},$$

with b and c formed from the balances of the tokens the swap does not touch. Balances are rescaled to eighteen decimals before either loop runs, because the invariant treats its tokens as interchangeable at par and a six-decimal stablecoin read on an eighteen-decimal scale is off by a factor of 10^{12}. Output is net of the pool fee,

$$\Delta_o = (R_{o,p,e^-} - R'_{o,p,e} - 1)\varphi_p,$$

carrying the contract's own off-by-one guard on the raw difference.

A_p is Curve-specific and the standardised subgraph schema drops it, so it is identified from data instead of fetched. Every other quantity in the invariant is observed and every Curve swap reports both amounts, which leaves A_p as the only unknown,

$$\widehat{A}_p = \arg\min_{A_p} Q_{0.90} \left\{ \frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(A_p)) - \Delta_o|}{\Delta_o} \right\},$$

fitted on the first half of a pool-day's trades over a logarithmic grid with a ternary refinement, then evaluated on the trades not used for estimation. A pool-day is retained when the 90th percentile of fitting error is below 1%. Using the upper tail prevents a partial fit from passing: one misspecified pool produced a median fitting error below 0.1% but a 34% median error on the other trades because the invariant fit only part of its activity. Curve has also operated crypto-pools since 2021 that follow the CryptoSwap invariant. Applying StableSwap to these pools yields a best-fit A_p of 3 and a 36% median quote error, so pool-days whose observed trades are inconsistent with StableSwap remain outside this component exercise.

Table 18 reports the held-out fit by validation day.

Across four validation days, held-out median errors range from 0.029% to 0.039%.

Table 18: StableSwap calibration and held-out quote error, by validation day

Day	Pools fitted	Pools excluded	Held-out trades	Median $\widehat{A}_p$	Median error	Within 1%
2021-05-28	9	14	367	12,716	0.030%	100.0%
2022-09-12	11	16	262	20,493	0.029%	100.0%
2023-12-28	20	28	471	23,428	0.039%	98.9%
2025-04-13	42	50	1,207	46,851	0.036%	99.7%

Note: The sample contains four sampled days. $\widehat{A}_p$ is reported in the module's convention, which carries the on-chain precision factor of 100. Median error is the median absolute deviation of the quote from realised output on trades the calibration never saw.

B.4 Weighted pools

Balancer weighted pools use a weighted geometric-mean invariant,[12]

$$K_{w,p} = \prod_j R_{j,p,e^-}^{w_{j,p}}, \qquad \sum_j w_{j,p} = 1,$$

which gives the two-token exact-input form

$$\Delta_o = R_{o,p,e^-} \left[1 - \left(\frac{R_{i,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i} \right)^{w_{i,p}/w_{o,p}} \right].$$

Only the ratio $w_{i,p}/w_{o,p}$ enters. An equal-weight Balancer pool and a constant-product pool with the same reserves therefore return the same quote. Balancer's weighted-pool contract accepts inputs up to 30% of the input-token reserve; that contract limit governs whether a quote can execute, but it is not the empirical support rule.[13] The counterfactual design applies the common 5% own-price-impact cutoff in Equation (9) to Balancer and every other pool family. The subgraph reports the weight ratio $\theta_p = w_{i,p}/w_{o,p}$ and the swap fee at its head block while balances come from a historical snapshot, so a pool that has repriced itself since is quoted with a current parameter against a past state. Liquidity-bootstrapping pools move weights on a schedule by design and managed pools move them on demand, and a weighted pool's owner can reset the fee at will. What such a pool produces is a small and nearly constant relative bias, which appears directly in the preceding validation: reported parameters reproduce trades in 2024 to a millionth of a basis point and trades in 2022 only to 0.19%. A gap of that size is a fee. The fee enters the quote through $\varphi_p \Delta_i$, so a fee wrong by a basis point moves output by about a basis point, whereas a mistake in the invariant would leave an error that scales with trade size.

We therefore treat both as unknowns to be recovered from the trades a pool actually executed, the move Curve's amplification coefficient needed in Section B.3, and we keep a reported value only where it survives that test. Reconstructing a pool's state from its own event history follows Lehar and Parlour (2025); what the weighted family adds is that two arguments of the pricing function

[12] See Martinelli and Mushegian, *A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor*.

[13] See the input-ratio bound in Balancer's `WeightedMath.sol`.

are themselves missing from the historical record, so a reconstructed state alone does not deliver a quote. Every quantity in the weighted invariant other than θ_p and f_p is observed, and every Balancer swap reports both amounts, so with balances reconstructed either parameter is the only unknown. We order a pool-day's trades and assign them alternately to a fitting set $\mathcal{O}_p$ and an evaluation set, and every error reported in Table 19 is measured on the evaluation set. For a trial parameter pair ϑ,

$$\mathcal{E}_p(\vartheta) = Q_{0.90}\left\{\frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(\vartheta)) - \Delta_o|}{\Delta_o} : e \in \mathcal{O}_p\right\}$$

is the fitting error, defined only when at least nine tenths of $\mathcal{O}_p$ returns a quote at all and undefined otherwise. The upper tail does the gating for the reason it does at Curve, and the coverage condition stops a parameter that prices one corner of a pool's activity from standing for the pool.

Each token pair's weight ratio is fitted on the trades crossing it in either direction, with a trade running from b to a scored at the reciprocal exponent, which is what the invariant implies and which doubles the trades available to pin one number,

$$\widehat{\theta}_{ab,p} = \arg\min_{\theta} \mathcal{E}_p\big(\theta \text{ on } \mathcal{O}_p^{a\to b},\ \theta^{-1} \text{ on } \mathcal{O}_p^{b\to a}\big),$$

searched over a logarithmic grid spanning Balancer's own weight bounds and refined by ternary search in the log exponent, and attempted only for token pairs carrying at least four fitting trades, since one or two are fitted exactly by construction and then read as an exact fit. The fee $\widehat{f}_p$ is recovered the same way over the fee range the vault permits. Reading is tried before fitting and the fee before the weights, so each tier adds at most one free scalar and a pool that reproduces its trades on reported parameters is never handed a fitted one,

$$\widehat{\vartheta}_p = \begin{cases} (\theta_p, f_p) & \text{if } \mathcal{E}_p(\theta_p, f_p) \le \bar{e}, \\ (\theta_p, \widehat{f}_p) & \text{else if } \mathcal{E}_p(\theta_p, \widehat{f}_p) \le \bar{e}, \\ (\widehat{\theta}_{\cdot,p}, f_p) & \text{else if } \mathcal{E}_p(\widehat{\theta}_{\cdot,p}, f_p) \le \bar{e}, \\ \text{excluded} & \text{otherwise,} \end{cases} \qquad \bar{e} = 0.1\%. \tag{16}$$

The label a pool carries in the source plays no part in this. A pool-day whose observed trades no tier reproduces is excluded whatever its `poolType` says, and the third tier further requires the fitted ratios to cover nine tenths of the fitting trades, so a pool-day is never accepted on the token pairs that happened to fit and then quoted on the rest.

The frequency with which the parameter search binds measures the sufficiency of reported pool state. Of the 256 pool-days accepted across the twelve validation days, 209 clear the gate on reported parameters with nothing identified, 43 need the swap fee alone, and 4 need per-token-pair weight ratios. Four fifths of the priced sample therefore carries no fitted parameter, and identification repairs a stale field in a minority of pool-days instead of supplying the invariant.

The gate $\bar{e}$ is set at 0.1% and not at the 1% Curve uses, because the achieved errors separate

Table 19: Weighted-pool quoter and reserve-timing alternatives

Day	Held-out trades	Backward	Flat	Opening	Pools priced / excluded
2021-04-25	4	4.7×10^{-9}	3.00%	3.94%	1 / 0
2021-09-29	273	2.5×10^{-8}	1.24%	0.80%	18 / 8
2022-03-09	524	1.6×10^{-4}	1.05%	2.66%	25 / 4
2022-08-16	701	1.1×10^{-9}	0.75%	1.04%	33 / 12
2023-01-24	832	2.0×10^{-10}	3.73%	4.71%	44 / 11
2023-07-03	301	7.1×10^{-10}	1.78%	3.32%	26 / 19
2023-12-11	185	2.8×10^{-9}	1.77%	2.91%	17 / 20
2024-05-20	265	3.8×10^{-10}	4.81%	5.17%	15 / 36
2024-10-27	142	1.8×10^{-10}	1.68%	2.24%	12 / 36
2025-04-06	248	2.1×10^{-10}	5.32%	9.48%	19 / 40
2025-09-13	501	2.5×10^{-4}	1.09%	1.71%	27 / 25
2026-02-21	582	8.7×10^{-9}	0.68%	1.90%	19 / 6

Note: Median absolute quote error in percent of realised output, on trades no fit ever saw. *Backward* reconstructs per-swap balances by rolling the day's event sequence back from the closing snapshot, *Flat* keeps the snapshot balances constant across the day, and *Opening* treats the snapshot as the day's opening state and rolls forward. The sample contains twelve days from 2021-04 to 2026-02.

cleanly there. The threshold separates weighted pools, whose held-out errors range from 10^{-9} to 0.004%, from stable, composable-stable, and elliptic-curve pools, whose held-out errors range from 0.05% to 1.08% and have a 99th percentile of 12%. At a 1% gate the second group clears on a fitted weight ratio and arrives quoted instead of excluded, which is the Curve lesson in a second form: there the over-permissive object was a parameter range and here it is an error threshold.

Across the twelve date rows of Table 19, the Backward column returns a median error of 0.0000% and the Pools priced / excluded column identifies 256 of 473 testable pool-days as weighted pools. The other pool-days carry a median 91.8% of tested swap volume and follow different Balancer pricing rules. The exercise therefore validates the model for identified weighted pools, not for Balancer as a whole.

C Reconstructing pool balances before each trade

Every counterfactual quote uses the pool state immediately before the trade, whereas some historical records store balances only at the end of an hour or day. We recover the earlier state by ordering every balance-changing event within the period and unwinding the sequence from the recorded closing balance. The exercises below validate that procedure on selected constant-product and weighted-pool observations. The cost comparison retains only routes whose pools can be reconstructed in this way.

Let $\mathbf{R}_p^{\text{end}}$ be pool p's stored period-end reserve vector and $\boldsymbol{\Delta}_{p,e}$ the vector of net signed token amounts in event e. The state before event e is the stored value with every later event removed,

$$\mathbf{R}^{\text{pre}}_{p,e} = \mathbf{R}^{\text{end}}_{p} - \sum_{e' \geq e} \boldsymbol{\Delta}_{p,e'} = \mathbf{R}^{\text{pre}}_{p,e+1} - \boldsymbol{\Delta}_{p,e},$$

which is a single reverse pass over the period. The alternative reading takes the previous period's stored value as the current period's opening state and replays forward,

$$\widetilde{\mathbf{R}}^{\text{pre}}_{p,e} = \mathbf{R}^{\text{end}}_{p,\text{prev}} + \sum_{e' < e} \boldsymbol{\Delta}_{p,e'},$$

which accumulates every unrecorded balance change between the two stored points. Replaying forward returns a median absolute quote error of 11.8% against realised outputs, whereas unwinding backward returns 0.0000%.

Liquidity additions and withdrawals also move constant-product pool balances, so the ordered replay includes them alongside swaps. We unwind the sequence to the start of the period and compare the reconstructed opening balance with an independently observed reserve state,

$$\left\| \left(\mathbf{R}^{\text{pre}}_{p,1} - \mathbf{R}^{\text{end}}_{p,\text{prev}} \right) \oslash \mathbf{R}^{\text{end}}_{p,\text{prev}} \right\|_{\infty} < \tau,$$

Agreement establishes that the event history reproduces the observed reserves for the tested period. The state-dependent sample omits a period when a required event family is incomplete or the balances do not reconcile. Coverage is reported by exchange and year because missing state can be concentrated in actively managed or newly launched pools.

The same logic applies to weighted pools. We treat `poolSnapshots.amounts` as the closing balance and unwind joins, exits, and swaps in their recorded order. Table 19 compares this reconstruction with two alternatives that either hold the closing balance fixed or treat it as the opening state.

The Balancer replay includes joins and exits as well as swaps. Omitting those events assigns intraday liquidity changes to the pricing rule and changes the pool states available for the price comparison.

D Restricting hypothetical trades to observed size–depth combinations

Component-validation samples contain executed swaps and therefore pools deep enough to serve the observed trade. Hypothetical routes can demand much more output relative to pool depth, where the observed errors no longer describe pricing accuracy. We include a hypothetical leg in the cost comparison only when its own price impact is at most 5%, calculated under the pricing rule of the pool family it would traverse.

For constant-product swaps, input relative to reserves provides a transparent size–depth com-

Table 20: Size relative to depth in the validation population

Day	Swaps	Median	90th pct.	99th pct.	99.9th pct.	Maximum
2020-05-05	2	1.00%	1.00%	1.00%	1.00%	1.00%
2021-02-10	141,804	0.19%	2.00%	8.68%	23.17%	99.83%
2021-11-18	107,059	0.10%	2.24%	13.15%	71.64%	100.00%
2022-08-26	79,180	0.48%	3.66%	18.44%	94.13%	100.00%
2023-06-03	159,551	0.85%	4.93%	19.36%	96.59%	100.00%
2024-03-10	215,060	0.41%	3.61%	15.11%	57.81%	100.00%
2024-12-16	132,508	0.31%	3.05%	13.25%	71.93%	100.00%
2025-09-23	97,106	0.24%	2.51%	19.92%	99.11%	100.00%
Pooled	932,270	0.34%	3.29%	14.86%	78.33%	100.00%

Note: Trade size as a percentage of the input-side reserve of the pool traversed, over the constant-product swaps the quoters were validated against. The sample contains eight dates from 2020-05-05 through 2025-09-23.

parison,

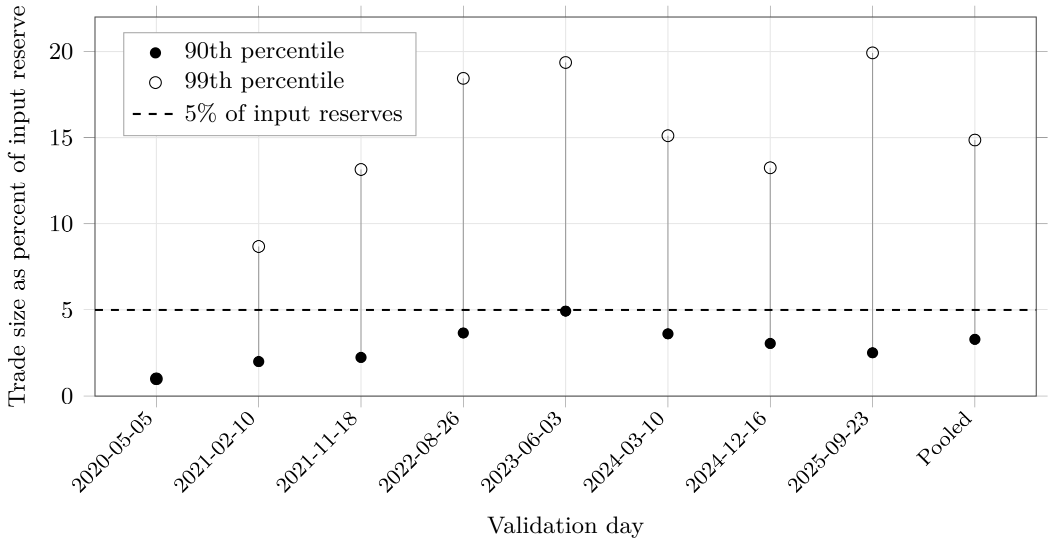
$$\iota_{i,p,e^-} = \frac{\Delta_i}{R_{i,p,e^-}},$$

the input amount as a fraction of the input-side reserve. Table 20 reports its distribution in the historical validation sample. The statistic helps calibrate what constitutes a large trade in a constant-product pool, but it does not replace the common own-price-impact rule for concentrated-liquidity, StableSwap, or weighted pools.

In the Pooled row of Table 20, five per cent of input reserves sits between the 90th percentile of 3.29% and the 99th of 14.86%; the same cutoff lies between those columns on seven of the eight date rows. Figure 5 shows the daily distributions. This calculation locates the cutoff in the empirical size distribution. The empirical rule remains a 5% own-price-impact cutoff computed separately under each pool family's pricing function.

Equation (9) applies the same 5% own-price-impact ceiling to every family. The component exercises establish pricing accuracy for realised swaps within their stated samples; they do not measure quote error on unexecuted routes.

Figure 5: Constant-product trade size relative to reserves



Note: The statistic is a leg's input amount as a fraction of the input-side reserve in the constant-product pool it traverses. Points mark the 90th and 99th percentiles; the dashed line marks 5% of reserves. The two percentiles coincide on 2020-05-05. The sample contains 932,270 realised swaps on eight dates from 2020-05-05 through 2025-09-23.

E Coverage of the price comparison

Omitting an exchange can change the best direct route, the best intermediated route, or both. Table 21 first describes the volume recorded in nine local source series, including Uniswap v1 as a separate architecture sample and the available Fluid extracts. Volume in the Graph-indexed series is reconstructed from individual swaps using the smaller dollar value of the two legs. This avoids double counting in two series and removes implausible oracle values in thin pools: one Curve pool reports a $456m day whose swaps sum to $8, and a constant-product pool with $255 of reserves reports $1.36bn. Physical bounds on volume relative to reserves exclude between 0.10% and 0.24% of Graph-indexed pool-days per year. Uniswap v1 is excluded from the route-cost comparison because its forced-routing architecture is studied separately. Fluid supplies volume on only 40 of the 400 sampled dates and no historical TVL field. The table consequently reports source coverage. Complete market shares require a common historical state series for every exchange.

The Uniswap v4 pricing model covers static-fee pools without hooks. Pool-identity reconstruction leaves zero swaps with incomplete or invalid immutable pool attributes across 29,005,863 rows. Pools in this group account for 84.02% of swaps and 99.05% of reported value; the corresponding 2026 shares are 79.89% and 95.57%. Other pools use hooks, dynamic fees, or both, whose transaction-level cash flows are not identified from the raw event. These shares describe the reach of the pricing model. They do not establish the availability of every pre-transaction state.

Curve carries between 9.68% and 18.70% of observed source volume in every year it exists and ranks second in four years, making its coverage economically important. The StableSwap model fits

Table 21: Share of observed decentralised-exchange volume, by source and year

Year	Uni V1	Uni V2	Uni V3	Uni V4	Sushi V2	Sushi V3	Curve	Balancer	Fluid
2020	2.09	75.89	0.00	0.00	10.89	0.00	11.12	0.00	0.00
2021	0.02	31.27	37.97	0.00	17.31	0.00	12.25	1.18	0.00
2022	0.01	7.01	64.07	0.00	4.35	0.00	18.70	5.87	0.00
2023	0.01	9.26	66.81	0.00	1.21	0.03	13.88	8.80	0.00
2024	0.00	11.63	69.06	0.00	0.49	0.01	12.57	5.86	0.38
2025	0.01	4.21	53.10	19.53	0.28	0.01	9.68	1.49	11.70
2026	0.01	2.27	46.02	31.94	0.11	0.16	12.61	0.15	6.73
Pooled	0.05	14.58	54.54	5.30	5.73	0.02	13.37	3.82	2.61

Note: Percentage shares of volume recorded in nine local source series. The weekly calendar contains 400 dates; Fluid is observed on 40 (6 in 2024, 25 in 2025, and 9 in 2026) and supplies no TVL field. Its entries are partial observed-volume contributions, not complete market shares or turnover estimates. The table displays 2020–2026 and pools volume over those years. Uniswap v1 is shown as the separate forced-routing architecture sample and is excluded from the route-cost opportunity set.

621 pool-days and excludes 588 whose 90th-percentile fitting error exceeds 1%. Across 28 sampled dates from 2020-02-11 to 2026-04-28, these exclusions account for 34.2% of the Curve volume assessed, or \$1.83bn of \$5.36bn. Another 2.75% of Curve volume occurs in pool-days with too few trades to estimate and evaluate the amplification coefficient separately. Lowering the minimum from eight trades to four changes annual excluded shares by at most 1.3 pp. Table 22 shows where these exclusions occur.

The exclusions are concentrated in routes involving the native asset. In Panel B of Table 22, the excluded share of native-leg volume exceeds the stable-leg share by at least 33 pp in every sample year and by 54 pp in the 2023 row. Tricrypto pools pairing WBTC and WETH with a stablecoin account for \$818m, or 44.6%, of the \$1.83bn excluded, including seven of the twelve largest excluded pool-days. These pools use CryptoSwap, so the missing coverage is concentrated where Curve supplies depth between ETH and stablecoins.

SushiSwap v3 is omitted from the route-cost comparison because its historical records lack the pool's root price, in-range liquidity, and tick spacing. Its reserves are distributed across ticks, and the balance and weight fields exposed by the common schema do not describe that invariant. The exchange accounts for 0.016% of volume among the seven exchanges used for route costs over the full sample and 0.176% in 2026. Across fourteen sampled days, only 4.1% of its volume occurs on token pairs unavailable on another included exchange, never more than \$37,609 on one day.

Balancer accounts for 3.9% of measured venue volume over the full sample and 8.8% at its 2023 peak. Table 19 validates selected weighted pools. Linear and boosted pools remain outside the quoter. Of 29 sampled composable-stable pools, 23 reduce to two external tokens after removing the pool's own share token, but pricing them also requires historical rate-provider states absent from the source. The priced Balancer sample is therefore confined to weighted pools whose historical state can be reconstructed and whose realised trades are reproduced at one of the tiers of Equation (16).

Table 22: Curve volume outside the StableSwap comparison

Panel A: composition of the exclusion					
Leg served	Pool-days	Volume	Of excluded	Of priced	Of that leg
Native	374	\$1.038bn	56.6%	15.7%	65.2%
Stable	119	\$0.752bn	41.0%	79.9%	21.1%
Other volatile	95	\$0.045bn	2.4%	4.4%	22.2%
Panel B: excluded share of each leg's volume, by year					
Year		Native leg		Stable leg	Other volatile
2021		85.49%		46.91%	15.23%
2022		68.82%		30.21%	50.77%
2023		59.11%		5.51%	56.60%
2024		48.24%		0.94%	14.38%
2025		47.47%		13.96%	10.52%
2026		60.92%		9.04%	20.08%

Note: A pool serves the *native* leg when it contains ETH or a staking derivative, the *stable* leg when every token is stable, including baskets and interest-bearing wrappers, and the *other volatile* leg otherwise. Stablecoin base-pool tokens are classified with the underlying stable assets. The sample contains 28 measured Curve pool-days from 11 February 2020 through 28 April 2026.

F Round-trip exclusion

A route whose first input token equals its last output token,

$$\mathbf{1}\{\, k_0 = k_{m(r)} \,\},$$

does not represent a conversion between distinct endpoint assets. We exclude these routes from the dominance measures. The excluded population can contain cyclic arbitrage and other self-returning activity; the endpoint rule does not classify every route as arbitrage or wash trading. Multi-trade arbitrage can be bundled atomically (Daian et al., 2020), but identifying trader intent would require additional evidence.

Table 23 reports the distribution across sampled days; Figure 6 shows the day-level observations.

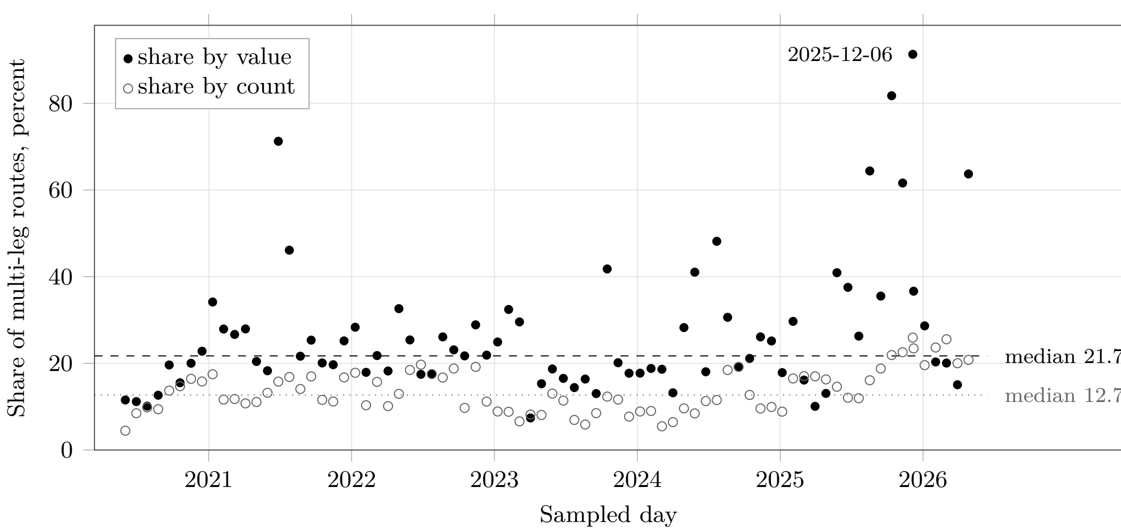
Across the quantile rows of Table 23, the Share by value column exceeds Share by count by a factor between 1.7 and 3.5, consistent with a population in which large self-returning routes carry disproportionate value. Round trips appear on every sampled day; their minimum count share is 4.45%. The paper excludes them before forming both count- and value-weighted dominance measures. Figure 6 plots their count and value shares among multi-leg routes and identifies the date of the maximum.

Table 23: Round-trip share of multi-leg routing, across 79 sampled days

Quantile across days	Share by count	Share by value
Minimum	4.45%	7.40%
25th percentile	9.60%	17.73%
Median	12.70%	21.72%
75th percentile	17.04%	29.56%
90th percentile	20.03%	46.11%
Maximum	25.93%	91.31%
2020 median	11.77%	14.08%
2021 median	13.21%	25.35%
2022 median	16.72%	21.89%
2023 median	8.51%	17.71%
2024 median	9.60%	21.13%
2025 median	16.75%	36.08%
2026 median	20.80%	20.32%

Note: Round trips as a share of multi-leg routes. The sample contains 79 dates from 1 June 2020 through 27 April 2026 and 13,160,047 routes, of which 2,488,399 are multi-leg and 351,553 are round trips. Dates with fewer than 200 multi-leg routes are omitted.

Figure 6: Round-trip contamination of multi-leg routing, day by day



Note: Each mark is one sampled date. *Share by value* weights round-trip routes by their notional; *share by count* gives each route equal weight. Both shares use multi-leg routes as the denominator. Dashed and dotted lines mark the respective medians, and the highest point is labelled. Dates with fewer than 200 multi-leg routes are omitted. The sample contains 79 dates from 1 June 2020 through 27 April 2026 and 2,488,399 multi-leg routes.

G Additional liquidity-provider estimates

Table 24 reports the vehicle-token flow outcomes that accompany the V4 stock, action, and range estimates in Table 15. Tables 25, 26, and 27 report every horizon and outcome underlying the main-text protocol comparison. These estimates retain the same vehicle-date comparison and observational interpretation as Equation (15).

Table 24: Stablecoin shortfalls, V4 flash accounting, and LP-flow sides

Interaction term ($G^S_{c,t} \times M_{c,t}$)	Gross LP flow [log points]	Add-side LP flow [log points]	Remove-side LP flow [log points]
Stable gap × internal same-asset share	-0.327^{***} (0.057)	-0.303^{***} (0.057)	-0.354^{***} (0.061)
Stable gap × multi-leg transaction share	-0.073^{**} (0.035)	-0.097^{***} (0.034)	-0.048 (0.037)
Stable gap × gross-to-net reduction share	$+0.214^{**}$ (0.090)	$+0.156^{*}$ (0.090)	$+0.276^{***}$ (0.095)
Observations / date clusters			1,209 / 403
Asset and date effects			Yes
Origin-day activity controls			Yes

Note: Each cell reports the interaction of a 10 pp larger stablecoin route-minus-capital gap with a 10 pp larger V4 accounting proxy. Outcomes are future log admissible vehicle-token-side modify-liquidity flow over 120 days. The unit, fixed effects, controls, covariance, and stablecoin sample match Table 15. Add and remove flows value the vehicle-token sides assigned through V4 modify-liquidity records. These predictive specifications measure vehicle-token flows; whole-pool inventory and causal LP supply remain outside their scope. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 25: Stablecoin shortfalls and the V4-minus-V3 LP-action response

Outcome [log points]	7 days	30 days	120 days
LP actions	+0.163***	+0.212***	+0.320***
	(0.016)	(0.015)	(0.013)
Active origins	+0.045***	+0.062***	+0.115***
	(0.011)	(0.011)	(0.010)
Stacked observations / dates			5,198 / 522
Asset-date and protocol effects			Yes
Current protocol LP controls			Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 pp larger stablecoin route-minus-capital gap and the future LP-action outcome, with standard errors in parentheses. Outcomes are log points. The stacked unit is a protocol–vehicle–date row restricted to dates with observed V4 singleton swap activity. All columns absorb vehicle-date and protocol fixed effects, include same-protocol current LP-action and active-origin controls, and cluster by date. V3 outcomes are mint and burn event counts and active origins; V4 outcomes are modify-liquidity event counts and active origins. The estimates are conditional protocol differences and do not identify the effect of V4 adoption. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 26: Stablecoin shortfalls and the V4-minus-V3 LP-flow response

Outcome	7 days	30 days	120 days
Gross LP flow [log points]	−0.043	−0.009	+0.029
	(0.033)	(0.027)	(0.022)
Add-side flow [log points]	−0.043	−0.023	+0.012
	(0.033)	(0.027)	(0.023)
Remove-side flow [log points]	−0.032	+0.007	+0.046**
	(0.034)	(0.028)	(0.022)
Narrow/medium flow share [pp]	+1.530***	+1.126***	+0.329***
	(0.236)	(0.133)	(0.095)
Broad flow share [pp]	−1.530***	−1.126***	−0.329***
	(0.236)	(0.133)	(0.095)
Stacked observations / dates			5,188 / 521
Asset-date and protocol effects			Yes
Current protocol LP-flow controls			Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 pp larger stablecoin route-minus-capital gap and future vehicle-token-side LP flow, with standard errors in parentheses. Flow-amount outcomes are log points; range-composition outcomes are pp. The stacked unit is a protocol–vehicle–date row restricted to dates with observed V4 singleton swap activity. All columns absorb vehicle-date and protocol fixed effects, include same-protocol current gross, add, remove, and sender-day flow controls, and cluster by date. V3 outcomes value mint and burn token sides; V4 outcomes value modify-liquidity token sides. The estimates are conditional protocol differences and do not identify the effect of V4 adoption. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 27: Stablecoin shortfalls and the V4-minus-V3 reported-liquidity footprint

Horizon	Reported TVL growth [log points]	Pool-footprint growth [log points]
7 days	$+0.237^{***}$	$+0.075^{***}$
	(0.073)	(0.017)
30 days	$+0.297^{***}$	$+0.149^{***}$
	(0.056)	(0.020)
120 days	$+0.637^{***}$	$+0.275^{***}$
	(0.042)	(0.022)
Obs. / date clusters (7d)		2,986 / 301
Obs. / date clusters (30d)		2,756 / 278
Obs. / date clusters (120d)		1,856 / 188
Asset-date and protocol effects		Yes
Origin TVL and pool-count controls		Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 pp larger stablecoin route-minus-capital gap and future growth in vehicle-linked reported full-pool TVL or pool count, with standard errors in parentheses. Outcomes are changes in log one plus the reported quantity. The stacked unit is a protocol–vehicle–date row restricted to V4-active dates with strict origin and target support in both protocols. All columns absorb vehicle-date and protocol fixed effects, include same-protocol origin TVL and pool-count controls, and cluster by date. A pool contributes its full reported TVL when it contains the vehicle; vehicle–vehicle pools can enter both vehicle series. The quantity is a reported stock-and-footprint proxy; reconstructed deposited capital, side-specific LP inventory, and causal protocol effects remain outside its scope. Asterisks $*$, $**$, and $***$ denote statistical significance at the 10%, 5%, and 1% levels, respectively.

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